Two-Stage Robust Differential Evolution for Multimodal Optimization

***Abstract*—Multimodal optimization (MMO) locates all global optima of a multi-peaked landscape and treats them as equally desirable. Many engineering applications instead need solutions that remain near-optimal under decision-variable perturbation, which singles out the wide-basin optima among peaks of equal height. This paper studies the resulting robust multimodal optimization problem (RMOP) and contributes three coordinated components. *RobBench* is a benchmark suite whose robustness ground truth is derived analytically from the Gaussian peak-width parameters; its 20 problems span dimensionality, robustness ratio, peak multiplicity, anisotropy, and spatial traps. *RobF1* is a metric that jointly penalizes redundant and missed robust-peak recommendations. *TSRDE* is a two-stage differential evolution algorithm that separates multimodal exploration from robustness-aware selection, combining dimension-adaptive expected-fitness estimation with competitive elimination to return a user-specified number of robust solutions. On RobBench, TSRDE attains a mean RobF1 of 0.952, significantly above eight multimodal and robust baselines; its Phase-1 configuration also leads the same baselines on the independent CEC2026 MMO benchmark, indicating that robustness awareness does not come at the cost of multimodal discovery. The robustness evidence is drawn from equal-height Gaussian-mixture landscapes; validation on heterogeneous landscapes remains open. Source code: https://github.com/Wenhua-Li/TSRDE.**

***Index Terms*—Multimodal optimization, robust optimization, differential evolution, benchmark, evaluation metric, basin width**

I. Introduction

In many engineering design problems, the decision-maker is confronted with multiple competing optima of comparable quality. Identifying all of them matters when the practical criterion diverges from the nominal objective: a safety-critical application may favor a second-best configuration whose performance degrades gracefully under manufacturing tolerances over the global optimum that collapses under even a small perturbation. This situation arises in aerodynamic shape design [1], chemical reactor parameterization [2], and structural sizing problems [3], where the gap between nominal optimality and operational robustness can be decisive. Fig. 1 illustrates this contrast on a two-peak landscape.

The multimodal optimization (MMO) community has developed a rich family of niching methods for multi-optima problems [4]–[6]: crowding-based approaches maintain diversity by replacing similar individuals during selection [7]; species-based methods partition the population into clusters, each converging to a distinct optimum [8], [9]; archive-guided strategies store found optima and redirect the search [10]; and surrogate-assisted variants handle expensive fitness evaluations by building approximate landscape models [2], [3], [11].

These methods target a well-defined problem: locate all global optima as precisely and completely as possible. We instead address a complementary setting that arises in engineering practice, where the decision-maker needs not all optima but a shortlist of solutions drawn from optima of equal nominal quality that remain near-optimal under decision-variable perturbations. In this setting, basin width, the spatial extent of the region within which a solution maintains near-optimal performance, varies substantially across optima and determines their practical value [12]. Robust single-objective optimization [13]–[15] provides variance-based and worst-case measures of this sensitivity, but addresses a single optimum. Combining robustness assessment with the multi-optima discovery machinery of MMO has received limited attention.

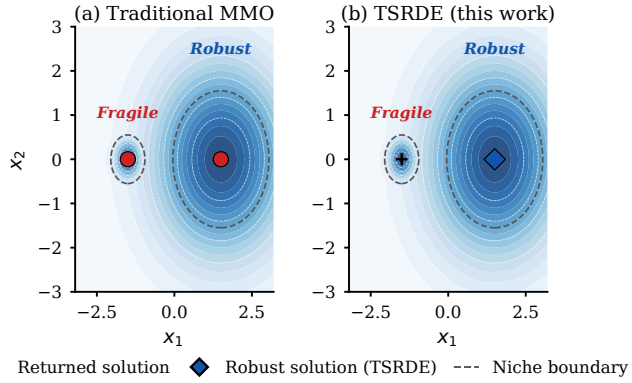


Fig. 1. Research motivation. (a) Traditional MMO returns all peaks regardless of basin width, including the fragile narrow peak. (b) TSRDE returns only the wide-basin robust peak, meeting the operational-stability requirement under perturbation.

The algorithm DREA [16] takes a step in this direction by coupling a peak-detection stage with a targeted robust search around each found peak. TSRDE instead defers all robustness evaluation to a second stage that selects among the found peaks, concentrating the robustness budget on discriminating robust from non-robust basins; the full comparison is developed in Section IV. Other recent work addresses related but distinct problems, including multimodal multiobjective optimization [17], [18], surrogate-assisted multimodal search [2], [11], and robust multi-objective optimization [19]–[21]. None provides the combination of explicit robustness annotation, a dedicated evaluation metric, and an algorithm designed specifically for the RMOP.

Closing this gap requires, above all, a precise problem statement, which Section II provides as the robust multimodal optimization problem (RMOP). It then takes three coordinated components: a *benchmark* that encodes basin-width variation explicitly so that robust peak identification is measurable (existing benchmarks [22], [23] annotate peak locations and function values but not robustness properties); a *metric* that

rewards recommending robust peaks while penalizing redundant recommendations and missed peaks (standard metrics, such as success rate and peak ratio [24], treat all global optima symmetrically); and an *algorithm* capable of covering multiple basins efficiently while directing its robustness evaluation budget toward targeted discrimination. The contributions of this work can be summarized as follows:

1) **RobBench**, a suite of 20 benchmark problems across five groups scaling dimensionality, robustness ratio, peak count, basin anisotropy, and spatial-trap structure. All peaks share the same nominal height, making basin width the sole differentiating factor, with robustness ground truth determined analytically from the function parameters.
2) **RobF1**, an F1-style metric that simultaneously penalizes two failure modes: redundant recommendations (multiple solutions assigned to the same robust peak) and omission (missed robust peaks).
3) **TSRDE** (Two-Stage Robust Differential Evolution), which decouples global multimodal exploration from robustness-aware refinement. Phase 1 builds a diverse candidate archive via k-NN rival-crowding DE with stagnation-triggered restarts, spending no evaluation budget on robustness assessment. Phase 2 estimates the expected fitness under local perturbation, $\hat{\mu}(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\delta}}[f(\mathbf{x}+\boldsymbol{\delta})]$, with a theoretically derived dimension-adaptive sample count $M^*(D)$, and applies a greedy diversity-preserving competitive elimination loop to return exactly n_{sol} robust solutions. All internal parameters are derived from the problem dimension and search range; n_{sol} is the only quantity the user must supply.

Experiments on RobBench show that TSRDE significantly outperforms all eight competing algorithms in RobF1 (mean 0.952; Wilcoxon signed-rank with Bonferroni correction). An independent evaluation on the CEC2026 MMO benchmark, in which TSRDE runs in conventional MMO mode without robustness-based selection, further shows that this robustness capability is not bought at the cost of peak-finding performance; its scope and limits are discussed in Section V.

II. Problem Formulation

Consider the continuous optimization problem

$$\min_{\mathbf{x}\in\Omega} f(\mathbf{x}), \quad \Omega \subseteq \mathbb{R}^D, \tag{1}$$

where $f : \mathbb{R}^D \to \mathbb{R}$ is the objective function and D is the dimensionality. In multimodal optimization, f admits multiple global optima $\mathcal{X}^* = \{\mathbf{x}_1^*, \ldots, \mathbf{x}_K^*\}$ with $K \geq 1$, all satisfying $f(\mathbf{x}_k^*) = f^*$ for $k = 1, \ldots, K$.

Traditional multimodal optimization algorithms aim to locate all members of $\mathcal{X}^*$ [4], [5]. In engineering practice, however, solutions of equal nominal quality can differ sharply in their sensitivity to decision-variable perturbations [12]: a solution on a sharp peak achieves $f(x) \approx f^*$ but degrades rapidly under manufacturing tolerances, whereas a solution in a flat basin maintains near-optimal performance across a neighborhood of perturbations. This distinction between *fragile* and *robust* optima is well recognized in single-objective robust optimization [14], [15] but has not been systematically incorporated into the multimodal setting.

A. Robustness Measure

We formalize the notion of basin quality through the expected objective value under local perturbation. Given a perturbation radius $\sigma_r > 0$, define the *expected fitness* of a candidate solution $\mathbf{x}$ as

$$\mu(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\delta}\sim U([-\sigma_r,\sigma_r]^D)}\big[f(\mathbf{x}+\boldsymbol{\delta})\big]. \tag{2}$$

A solution with low $\mu(\mathbf{x})$ remains near its nominal value across the perturbation neighborhood; a solution perched on a narrow peak sees $\mu(\mathbf{x}) \gg f(\mathbf{x})$ because the objective degrades rapidly in most perturbation directions.

This definition follows the expected-fitness paradigm established in robust single-objective optimization [1], [12] and adopted by recent robust multimodal algorithms [16]. Its key analytical advantage over variance-based measures becomes apparent in anisotropic landscapes: for a Gaussian peak with per-dimension widths $s_{k,1}, \ldots, s_{k,D}$, a Taylor expansion gives

$$\mu(\mathbf{x}_k^*) - f(\mathbf{x}_k^*) \approx \frac{h\sigma_r^2}{6} \sum_{d=1}^{D} s_{k,d}^{-2}, \tag{3}$$

where h is the peak height. The excess above the nominal value is a second-order average of the per-dimension widths; for isotropic peaks ($s_{k,d} \equiv s_k$) it is a strictly decreasing function of s_k, so the criterion reproduces the ground-truth ordering exactly. Define the *basin width* of a peak as $\sigma_k = \frac{1}{D}\sum_{d=1}^{D} s_{k,d}$. In general, the three natural summaries of the widths, namely the expected-fitness term $\sum_d s_{k,d}^{-2}$, the harmonic mean $D/\sum_d s_{k,d}^{-1}$, and the arithmetic mean σ_k, are *not* order-equivalent; RobBench's anisotropy is deliberately moderate, and we verified that on every one of the 20 problems the three criteria induce the identical full ranking, including the robust/non-robust boundary (smallest relative margin 2.3%; per-problem verification released with the code). The expected-fitness criterion is therefore consistent with the benchmark's ground truth on this suite. A variance-based measure, by contrast, involves a fourth-order term $\sum_d s_{k,d}^{-4}$, which is dominated by the single narrowest dimension; it can therefore rank a peak that is wide in all but one direction below a genuinely wide-basin peak, the failure mode that motivates the expected-fitness choice.

Definition 1 (Robust Peak)**.** *Peaks are ranked by basin width σ_k in descending order, and the top n_{rob} are designated* robust*. The robust set is denoted $\mathcal{R} \subseteq \mathcal{X}^*$ with $|\mathcal{R}| = n_{rob}$.*

B. Robust Multimodal Optimization Problem

The robust multimodal optimization problem (RMOP) seeks a solution set $S = \{x_1, \ldots, x_{n_{\text{sol}}}\}$ satisfying the following three conditions simultaneously:

1) **Near-optimality**: Each $x_i \in S$ satisfies $f(x_i) - f^* < \varepsilon$, ensuring all returned solutions are competitive in the nominal sense.
2) **Robustness coverage**: Each x_i covers a distinct robust peak in $\mathcal{R}$, meaning that the algorithm must discover and

select wide-basin optima rather than simply locating the nearest nominal optimum.

3) **Spatial diversity**: No two solutions are within the niche radius of the same peak, preventing the algorithm from returning redundant recommendations of the same basin.

The parameter n_{sol} is the number of robust solutions the user requests. In the ideal case $n_{\text{sol}} = n_{\text{rob}}$, but the user will typically not know n_{rob} exactly; Section V-E examines the sensitivity to this mismatch.

An important subtlety of the RMOP is that conditions (1) and (2) can conflict. Because all peaks share the same nominal height in RobBench, an algorithm cannot rely on fitness alone to identify robust candidates; it must explicitly evaluate how the objective behaves under perturbation. This is the core difficulty that separates RMOP from both standard MMO (which ignores condition (2)) and single-objective robust optimization (which ignores condition (3)).

III. RobBench: A Benchmark Suite for Robust Solution Discovery

A. Design Principles and Function Family

To evaluate algorithms on robust solution discovery, we require benchmarks with the following characteristics:

1) **Analytical ground truth**: robustness is encoded directly in the function parameters, eliminating the need for numerical estimation.
2) **Equal-height peaks**: all peaks achieve the same objective value $f^* = -10$, isolating basin width as the sole discriminant.
3) **Controlled difficulty axes**: five problem groups (A–E) vary dimensionality, robustness ratio, number of peaks, anisotropy, and spatial distribution independently.
4) **Calibrated separation**: the narrow/wide peak width ratio is set to 5:1 ($\sigma \in [0.30, 1.50]$), after pilot experiments showed that the original 12:1 ratio (v1: $\sigma \in [0.10, 1.20]$) made traditional multimodal algorithms struggle even on basic exploration.
5) **Scalability**: problems span $D \in \{2, 3, 5, 10\}$, peak counts $K \in \{2, 4, 6, 8\}$, and robustness ratios $n_{\text{rob}}/K \in \{0.125, 0.25, 0.33, 0.50, 0.75\}$.

The Gaussian peak family is chosen because it makes these properties measurable. Any smooth peak is locally Gaussian in the vicinity of its optimum, since the second-order Taylor expansion at an interior optimum is exactly an anisotropic Gaussian; the family therefore models the generic local shape of engineering optima while remaining simple enough that the expected fitness under perturbation admits the closed form of (3). Basin widths $s_{k,d}$ are function parameters, so robustness ground truth requires no numerical estimation or simulation. Existing multimodal benchmarks [22], [23] instead construct peaks by composition, transformation, and rotation; their ground truth specifies *where* the optima are and *how good* they are, but not *how wide* their basins are, which is precisely the annotation robust solution discovery requires. The annotation also makes reported scores unambiguous: a RobF1 of 0.93 certifies precision and recall in robust-peak identification, whereas the same value under peak-based metrics could equally arise from sharp basins that no robustness criterion would endorse.

B. Function Definition

Each RobBench problem is a negative Gaussian mixture:

$$f(x) = \min_{k=1,\ldots,K} \left\{ -h \cdot \exp\left(-\frac{1}{2} \sum_{d=1}^{D} \frac{(x_d - c_{k,d})^2}{s_{k,d}^2} \right) \right\}, \quad (4)$$

where $h = 10$ is the peak height (constant across all peaks), $c_k \in \mathbb{R}^D$ is the center of peak k, and $s_{k,d} > 0$ controls the basin width along dimension d. For isotropic peaks, $s_{k,d} = \sigma_k$ for all d; anisotropic peaks use dimension-specific widths. The robustness of peak k follows Definition 1: peaks are ranked by σ_k in descending order, and the top n_{rob} are designated robust.

C. Problem Groups

RobBench comprises 20 problems organized into five groups (Table I). Fig. 2 shows representative 2-D landscapes from each group.

TABLE I
RobBench problem specification (20 problems, five groups). All peaks share the height $f^* = -10$; widths span $\sigma \in [0.30, 1.50]$ (isotropic groups) and $[0.30, 1.20]$ (Group D). Random selection achieves expected recall n_{rob}/K.

Group	Problem	D	K	n_{rob}
A	RB-A1	2	4	2
	RB-A2	3	4	2
	RB-A3	5	4	2
	RB-A4	10	4	2
B	RB-B1	5	4	1
	RB-B2	5	6	2
	RB-B3	5	4	2
	RB-B4	5	4	3
C	RB-C1	5	2	1
	RB-C2	5	4	2
	RB-C3	5	6	3
	RB-C4	5	8	4
D	RB-D1	5	4	2
	RB-D2	10	4	2
	RB-D3	5	4	2
	RB-D4	10	6	3
E	RB-E1	5	8	1
	RB-E2	10	8	2
	RB-E3	5	6	1
	RB-E4	10	6	2

Group A scales dimensionality from $D = 2$ to $D = 10$ while holding $K = 4$ and $n_{\text{rob}} = 2$ constant, testing whether robust peak discrimination survives an expanding search space.

Group B varies the robustness ratio n_{rob}/K from 0.25 (RB-B1: only 1 of 4 peaks robust) to 0.75 (RB-B4: 3 in 4), testing discrimination across robustness fractions.

Group C increases the total number of peaks from $K = 2$ to $K = 8$ at $n_{\text{rob}} = K/2$, stressing archive capacity and refinement budget allocation.

Group D introduces anisotropic peaks whose width varies across dimensions (e.g., one robust peak of RB-D1 has widths $[0.80, \ldots, 0.80, 0.30]$, giving $\sigma_k = 0.70$), testing the handling of non-spherical basins.

Group E presents spatial traps: several narrow peaks cluster in a high-density region near the origin while one or two

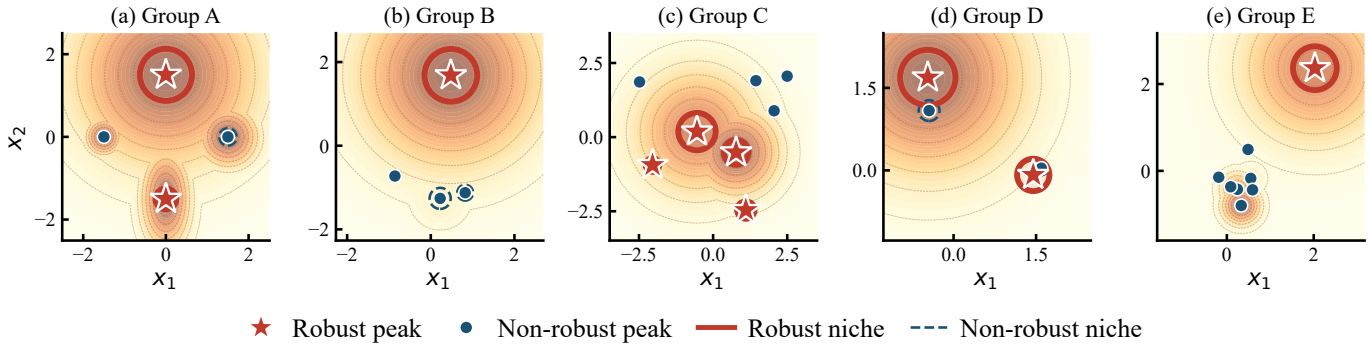


Fig. 2. Representative 2-D landscapes from each RobBench group. Stars (★) and blue dashed circles mark robust peaks; filled circles and red dashed circles mark non-robust peaks. (a) Group A: dimensionality; (b) Group B: robustness ratio; (c) Group C: peak count; (d) Group D: anisotropy; (e) Group E: spatial trap.

wide peaks sit isolated in remote corners (RB-E1: seven $\sigma = 0.30$ peaks near the origin, one $\sigma = 1.20$ peak in a corner). Traditional niche-based methods concentrate on the dense cluster; robust-aware algorithms must resist this local attraction.

D. Evaluation Metric for Robust Solution Discovery

Standard metrics like success rate, peak ratio (PR), and GNRS [24], measure coverage of all global optima without distinguishing robust from non-robust peaks; an algorithm locating all eight peaks in RB-E1 achieves PR= 1.0 yet only one peak is robust.

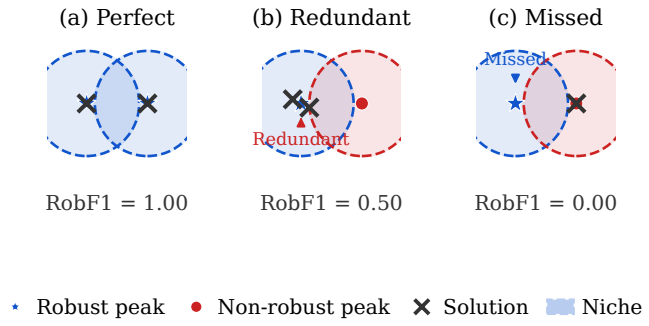


Fig. 3. Illustration of three RobF1 scoring scenarios. (a) Perfect score (1.0): each returned solution covers a distinct robust peak. (b) Redundancy (0.50): two solutions lie in the same robust niche, penalizing P_{rob}. (c) Missed robust peak (0.0): the only solution converges to a non-robust peak.

RobF1 addresses this gap. It is an F1-style metric that penalizes both redundant recommendations (multiple solutions covering the same robust peak) and missed robust peaks; the three scoring scenarios it distinguishes are illustrated in Fig. 3. Formally, given a solution set $S = \{x_1, \ldots, x_{n_{\text{sol}}}\}$, let $\rho_k = \frac{1}{2} \min_{j \neq k} \|c_k - c_j\|_2$ be the niche radius of peak k. Let n_{in} be the number of solutions lying within any peak's niche, and n_{cov} the number of distinct robust peaks covered by at least one such solution (de-duplicated). Robust precision and recall are:

$$P_{\text{rob}} = \begin{cases} n_{\text{cov}}/n_{\text{in}}, & n_{\text{in}} > 0, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

$$R_{\text{rob}} = n_{\text{cov}}/n_{\text{rob}}. \tag{6}$$

RobF1 is their harmonic mean:

$$\text{RobF1} = \begin{cases} \frac{2P_{\text{rob}}R_{\text{rob}}}{P_{\text{rob}}+R_{\text{rob}}}, & P_{\text{rob}} + R_{\text{rob}} > 0, \\ 0, & \text{otherwise}. \end{cases} \tag{7}$$

Proposition 1 (Perfect Score)**.** *RobF1 $= 1$ if and only if $n_{in} = n_{cov} = n_{rob}$, i.e., every in-niche solution covers a distinct robust peak and all robust peaks are covered.*

Proof. $P_{\text{rob}} \leq 1$ and $R_{\text{rob}} \leq 1$ hold by construction, and the harmonic mean of two quantities in $[0, 1]$ equals 1 if and only if both equal 1; $P_{\text{rob}} = 1$ requires $n_{\text{in}} = n_{\text{cov}}$, and $R_{\text{rob}} = 1$ requires $n_{\text{cov}} = n_{\text{rob}}$. □

When $n_{\text{rob}} = K$, RobF1 degenerates to a standard peak-level F1 score; when $n_{\text{rob}} < K$ it credits only the robust subset of $\mathcal{X}^*$. Two properties of the precision denominator deserve note. Counting only in-niche solutions (n_{in}) rather than all returned solutions avoids double-penalizing an algorithm for incomplete convergence, which R_{rob} already captures. Conversely, a solution converged to a *non-robust* peak increments n_{in} but not n_{cov}, diluting precision; recommending narrow peaks is precisely the failure mode RobF1 is designed to expose.

A deliberate scope choice of RobF1 should be made explicit: it measures the *coverage* conditions of the RMOP (conditions 2 and 3 of Section II) but does not itself verify near-optimality (condition 1), because niche membership is a spatial criterion. Near-optimality is instead enforced algorithmically by the Phase 2 quality screen and the local refinement that drives candidates to their basin floors, and it can be audited separately. We performed this audit post hoc on every solution returned in every RobBench run of all nine algorithms (audit script and output released with the code): all 385 solutions returned by TSRDE satisfy $f(x) - f^* < 10^{-13}$ (maximum observed excess 4.6×10^{-14}), whereas DREA and FastNEP return solutions with median excess 0.23 and 1.27, respectively, a distinction RobF1 alone would not expose.

IV. The TSRDE Algorithm

A. Motivation and Design Rationale

Traditional multimodal algorithms, like NCDE [25], LIPS [26], and species-based DE [8], locate peaks efficiently but apply no robustness criterion during the search, treating all optima as equally desirable. Surrogate-assisted variants [2], [3], [11] reduce evaluation costs but likewise treat peaks symmetrically. Algorithms that integrate robustness evaluation throughout the search, such as DREA [16] with its continuous expected-fitness assessment of every archive member, pay a substantial cost that limits basin discovery. TSRDE resolves this dilemma through strict phase separation. Phase 1 performs *no* robustness evaluations and spends its entire budget on discovering and archiving as many distinct basins as possible. Phase 2 applies robustness evaluation exclusively to high-quality archive members, using the expected fitness $\mu(\mathbf{x})$ defined in (2) and competitive elimination to return exactly n_{sol} robust solutions.

This separation ensures that robustness assessment never competes with exploration for the evaluation budget, and that no evaluation is wasted on perturbation sampling for candidates that will ultimately be discarded, the systematic inefficiency observed in DREA. All internal parameters (population size, archive capacity, perturbation radius, sampling count) are derived automatically from the problem dimension D and the search range, and the same rules are applied unchanged in every experiment of this paper; the desired number of robust solutions, n_{sol}, is the only quantity the user must supply. Where a closed form is stated in the remainder of this section, it is the form active under the experimental settings of Section V; the complete expressions, including defensive bounds that remain inactive there, are listed in Table II and implemented verbatim in the released source code. The quantities that most directly shape the output carry sensitivity evidence as well: Section V-E sweeps the user parameter n_{sol}, Section IV-D audits the sample count $M^*(D)$ against a large-sample reference, and Section V-D quantifies the design at the component level. Algorithm 1 presents the complete TSRDE procedure.

Component provenance and relation to existing methods. TSRDE deliberately assembles proven components rather than proposing new low-level operators: the k-NN rival crowding descends from crowding DE [7] and its nearest-neighbor refinements [25]; stagnation-triggered restarts are a standard device in multimodal DE; the parameter adaptation follows JADE [27]; archiving of found optima is common practice in MMO; and the Phase 2 polishers (Nelder–Mead, L-BFGS-B) are classical local-search procedures. The contribution lies in how the components are coupled and scheduled: a budget partition that spends zero robustness evaluations in Phase 1, restricts robustness evaluation in Phase 2 to a screened candidate set (Section IV-C1), and reinvests the savings in repeated refine-and-eliminate rounds. The ablation study (Section V-D) makes this attribution concrete: disabling the elimination loop costs far more than disabling the restart mechanism, indicating that the performance derives from the coupling rather than from any individual component.

Separating exploration from robustness assessment is not without precedent, and the distinctions should be stated explicitly. In robust *multi-objective* optimization, Deb and Gupta [19] assessed robustness as a modular post-processing step applied to whatever a standard MOEA returns, and DREA [16] similarly couples peak detection with a subsequent robust refinement stage. In both precedents, robustness assessment is a fixed post-hoc filter. In TSRDE, the split is instead an integral *budget-scheduling* design: the Phase 2 share $\alpha(D)$ and the sample count $M^*(D)$ are derived from the dimension, the candidate pool entering robustness evaluation is reduced by quality screening, and the budget saved by deferring robustness is recycled into a competitive elimination loop that refines and re-ranks the surviving candidates until exactly n_{sol} remain. Robustness evaluation as a post-processing step is, in principle, applicable to the archive of any MMO algorithm; TSRDE's contribution is a concrete budget-feasible realization of this idea whose internal quantities require no user tuning.

Algorithm 1 TSRDE: Two-Stage Robust Differential Evolution

Require: Problem f, bounds $[\mathbf{l}, \mathbf{u}]$, budget B, n_{sol}
Ensure: n_{sol} robust solutions
$N_{\text{est}} \leftarrow \max(20, 5D)$, $R_s \leftarrow (\mathbf{u}_1 - \mathbf{l}_1)/(5\,N_{\text{est}}^{1/D})$
$s^* \leftarrow \lfloor B_1/(12\,\text{NP}) \rfloor$ {restart patience, in generations}
$\alpha \leftarrow \alpha(D)$ (Table II); $B_1 \leftarrow \alpha \cdot B$ {Phase-1 budget share}
NP $\leftarrow 4\,N_{\text{est}}$ {population size}
Initialize population P (size NP), archive $A \leftarrow \emptyset$ (capacity cap $= 6\,N_{\text{est}}$)
{**Phase 1: Global Multimodal Detection**}
while FE $< B_1$ **do**
  Generate trial vectors via k-NN rival crowding DE (Section IV-B1)
  Update P using rival replacement; insert locally converged solutions into A (dedup radius R_s)
  if archive size stagnant for s^* consecutive generations **and** FE $< 0.9\,B_1$ **and** restarts < 5 **then**
    Reinitialize P via Latin-hypercube-stratified sampling (Section IV-B2)
  end if
  Update JADE parameters μ_F, μ_{CR}
end while
{**Phase 2: Robust Refinement and Competitive Elimination**}
$C \leftarrow$ top-K candidates from A whose fitness passes the quality band $f \le f_{\text{best}} + \text{band}$ (Section IV-C1)
for each $c \in C$ **do**
  Refine c locally (Nelder-Mead if $D \le 5$, else L-BFGS-B)
  $\hat{\mu}(c) \leftarrow$ expected-fitness estimate via $M^*(D)$ samples (Section IV-D)
end for
$C \leftarrow$ `diverse_rob_select`$(C,\ n_{\text{pre}},\ R_s)$ {pre-cull to $n_{\text{pre}} > n_{\text{sol}}$}
while $|C| > n_{\text{sol}}$ **do**
  for each $c \in C$ **do**
    Refine c locally; update $\hat{\mu}(c)$
  end for
  Remove worst $\lceil 0.33\,|C| \rceil$ candidates via `diverse_rob_select`
end while
Final local refinement on surviving C; deduplicate at radius R_s
return C

B. Phase 1: Global Multimodal Detection

Phase 1 employs differential evolution with three mechanisms adapted from recent multimodal optimization research, combined to maximize archive diversity without consuming evaluations on robustness assessment.

1) k-NN Rival Crowding Replacement: Standard crowding DE [7] replaces an offspring's nearest neighbor, but this can disrupt individuals that have already converged to a peak. We adopt a rival-based strategy: a trial vector $\mathbf{x}'$ generated from target $\mathbf{x}_i$ does not compete with $\mathbf{x}_i$ directly. Instead, among the k nearest neighbors of $\mathbf{x}'$ (where k is the population share per estimated peak, $k = \lfloor \mathrm{NP}/N_{\mathrm{est}} \rfloor$, four throughout this paper, decayed linearly to two as the Phase 1 budget is consumed), we identify the *rival* through a two-key lexicographic rule: candidates are first ranked by local competition fitness $\mathrm{LCF}(j)$, and ties are broken by crowding distance $\mathrm{CD}(j)$. Formally,

$$\mathcal{J} = \underset{j \,\in\, \mathrm{kNN}(\mathbf{x}')}{\arg\max}\ \mathrm{LCF}(j), \qquad \mathrm{rival} = \underset{j \,\in\, \mathcal{J}}{\arg\min}\, \mathrm{CD}(j), \tag{8}$$

where $\mathrm{LCF}(j)$ counts how many of the $k_{\mathrm{lcf}} = \min(20,\, 3D)$ nearest neighbors of individual j have better fitness (the cap keeps the convergence test local), so $\mathrm{LCF}(j) = 0$ marks a local optimum within its neighborhood and a higher $\mathrm{LCF}(j)$ marks an individual more dominated by its neighbors. $\mathrm{CD}(j)$ is the standard crowding distance, averaged over dimensions from the normalized gaps between j's two neighbors along each coordinate (boundary individuals receive an infinite-equivalent contribution). The rival is thus the most crowded individual among those least locally optimal, and the trial replaces the rival only if $f(\mathbf{x}') < f(\mathrm{rival})$. The rule protects exploratory individuals (high LCF) and isolated individuals (high CD), concentrating replacement pressure on crowded, converged individuals, the least likely to discover new peaks.

2) Stagnation-Triggered Population Restart: A restart is triggered when the archive size has not changed for $s^* = \lfloor B_1/(12\,\mathrm{NP}) \rfloor$ consecutive generations, where B_1 is the Phase 1 budget. The patience window is therefore derived from the budget itself rather than tuned: a stall of s^* generations consumes $B_1/12$ evaluations, half of the $B_1/6$ share available to each of the at most five restarts, so a fully stalled run still admits several restart opportunities within Phase 1. Restarts are issued only while less than 90% of B_1 has been consumed, and at most five restarts are allowed. When triggered, the population is fully reinitialized via Latin-hypercube-stratified uniform sampling over $[\mathbf{l}, \mathbf{u}]$, independently of the current archive contents, giving Phase 1 repeated opportunities to escape a converged configuration and discover additional basins. Section V-D quantifies this mechanism against a variant with restarts disabled.

3) JADE Adaptive Mutation and Crossover: The mutation scale factor F and crossover rate CR are adapted using JADE [27]: after each generation, the F and CR values of trials that successfully replaced their rivals are recorded, and the means are updated as

$$\begin{aligned} \mu_F &\leftarrow (1-c)\,\mu_F + c\,\frac{\sum_i F_i^2}{\sum_i F_i}, \\ \mu_{CR} &\leftarrow (1-c)\,\mu_{CR} + c \cdot \mathrm{mean}(\{\mathrm{CR}_i\}), \end{aligned} \tag{9}$$

with learning rate $c = 0.1$ (Lehmer mean for F, arithmetic mean for CR; μ_F and μ_{CR} initialized at 0.5 and 0.9). Individual F values are then sampled from a Cauchy distribution centered at μ_F and CR values from a normal distribution centered at μ_{CR}, both truncated to valid ranges. Adaptive parameter control in DE substantially improves performance across diverse problem landscapes [28]–[30], particularly in multimodal settings where different regions of the search space favor different mutation strengths.

4) External Archive: Solutions with local competition fitness $\mathrm{LCF} = 0$ (indicating local convergence) are inserted into an external archive A. Deduplication enforces a minimum pairwise distance R_s: if a new solution is within R_s of an existing archive member, only the better one is retained. Archive capacity is capped at $6\,N_{\mathrm{est}}$; when full, the worst-fitness member is discarded.

C. Phase 2: Robust Refinement and Competitive Elimination

Phase 2 refines archive members into a small set of robust solutions. It proceeds in four steps.

1) Step 1: Quality Screening: Define a fitness band around the archive's best solution:

$$\mathrm{band} = \max\big(0.05\,(f_{P_{90}} - f_{\mathrm{best}}),\ 0.5 \cdot (f_{P_{50}} - f_{\mathrm{best}})\big), \tag{10}$$

where $f_{P_{50}}$ and $f_{P_{90}}$ are the 50th and 90th percentiles of archive fitnesses. Only solutions with $f \le f_{\mathrm{best}} + \mathrm{band}$ proceed; that is, solutions no worse than the midpoint between the archive's best fitness and its median, with a floor of 5% of the best-to-P_{90} range so that near-degenerate archives do not collapse the band. From these, the top K candidates are selected as initial candidates, with $K = \max(5\,n_{\mathrm{sol}},\, 15)$, capped by the number of band-passing solutions to avoid over-committing the refinement budget: five refinement slots per requested solution, floored at fifteen so that the elimination loop retains slack even for small n_{sol}. This focuses the refinement budget on high-quality basins and enforces the near-optimality condition of the RMOP before any robustness evaluation is spent.

2) Step 2: Initial Robustness Evaluation: Each candidate first undergoes local refinement using Nelder-Mead (for $D \le 5$) or L-BFGS-B (for $D > 5$), targeting the original objective $f(\mathbf{x})$. After refinement, robustness is quantified by the expected fitness under local perturbation:

$$\hat{\mu}(\mathbf{x}) = \frac{1}{M^*(D)} \sum_{i=1}^{M^*(D)} f(\mathbf{x} + \boldsymbol{\delta}_i), \quad \boldsymbol{\delta}_i \sim U([-\sigma_r, \sigma_r]^D), \tag{11}$$

where $\sigma_r = 0.1 \cdot R_s$ and $M^*(D)$ is the dimension-adaptive sample count derived in Section IV-D. The perturbation radius σ_r is tied to the archive deduplication radius R_s, which itself reflects the characteristic inter-peak spacing estimated from N_{est} and the search range. A candidate with low $\hat{\mu}(\mathbf{x})$ sits in a wide basin that maintains near-optimal values across the perturbation neighborhood; a candidate on a narrow peak has $\hat{\mu}(\mathbf{x}) \gg f(\mathbf{x})$ because most perturbed points fall onto the steep shoulders of the peak.

Ranking candidates by $\hat{\mu}(\mathbf{x})$ rather than by a variance estimate is what separates wide from narrow basins. As

established in Section II, the expected fitness ranks anisotropic peaks by a second-order average of their widths and induces the same ranking as the benchmark's ground truth on every problem, whereas a variance criterion is dominated by the single narrowest dimension (the failure mode observed on Group D problems in preliminary experiments).

From the initial candidates, a pre-cull retains $n_{\text{pre}} = 3\,n_{\text{sol}}$ candidates using the `diverse_rob_select` procedure (Section IV-C5). This culling step removes clearly non-robust candidates before the more expensive elimination loop.

3) Step 3: Competitive Elimination Loop: While $|C| > n_{\text{sol}}$:

1) Apply local refinement to all surviving candidates, allocating 35% of the currently remaining evaluation budget to the round, split evenly across candidates.
2) Re-estimate $\hat{\mu}(c)$ for each candidate, refreshing the robustness scores.
3) Remove the weakest $\lceil 0.33\,|C| \rceil$ candidates (at least one) as identified by `diverse_rob_select`, which jointly accounts for robustness ranking and spatial coverage.

The elimination proceeds in rounds, each devoting more budget to a progressively smaller candidate pool. This concentrates evaluation effort on the final surviving candidates, reducing estimation noise in the crucial last comparison.

4) Step 4: Final Refinement and Deduplication: After the loop terminates with exactly n_{sol} candidates, any remaining Phase 2 budget is used for a final deep refinement pass on each solution. A deduplication sweep then removes near-duplicates within radius R_s, ensuring the output is spatially non-redundant.

5) Diverse Robust Selection: The `diverse_rob_select`(C, n, R_s) procedure selects the best n candidates from C while balancing robustness quality and spatial diversity:

1) Sort candidates by $\hat{\mu}$ in ascending order (lowest expected fitness, i.e., most robust, first).
2) Initialize selected set $S \leftarrow \emptyset$.
3) For $i = 1, \ldots, n$, compute an effective score for each remaining candidate c and greedily select the minimum:

$$\text{score}(c) = \hat{\mu}(c) + \lambda \mathbb{1}\big[\min_{s \in S} \|\mathbf{x}_c - \mathbf{x}_s\| < R_s\big], \quad (12)$$

where λ is a penalty constant chosen larger than the spread of the current $\hat{\mu}$ scores, so that any solution that crowds an already-selected one is dominated by any non-crowding alternative.

The indicator term in (12) is activated only when c falls within the niche radius of an already-selected solution, demoting it below all spatially distinct candidates regardless of its robustness rank. This creates a hard-diversity guarantee: the procedure never returns two solutions from the same basin, even if both happen to have better robustness scores than candidates from other basins.

D. Dimension-Adaptive Sampling for Robustness Estimation

The sample count $M^*(D)$ in (11) must be large enough that the estimated expected fitness $\hat{\mu}(\mathbf{x})$ reliably ranks robust peaks above non-robust ones, yet small enough to keep the Phase 2 evaluation budget manageable. We derive $M^*(D)$ from two independent lower bounds.

The *statistical precision bound* $M_{\text{stat}}(D)$ keeps the relative standard error of $\hat{\mu}(\mathbf{x})$ below a threshold ε with confidence $1-\delta$. A kurtosis analysis of the perturbed-objective distribution near an isotropic peak yields the following closed form.

Proposition 2 (Excess kurtosis and precision bound)**.** *Near the optimum of an isotropic peak, the perturbation-induced excess $f(\mathbf{x}_k^* + \boldsymbol{\delta}) - f(\mathbf{x}_k^*)$, $\boldsymbol{\delta} \sim U([-\sigma_r, \sigma_r]^D)$, has excess kurtosis $-6/(7D)$, and the sample count that keeps the relative standard error of $\hat{\mu}$ at most ε with confidence $1-\delta$ is*

$$M_{stat}(D) = \left\lceil \frac{z_{\delta/2}^2 \cdot (2 - 6/(7D))}{\varepsilon^2} \right\rceil. \quad (13)$$

Proof. Near the optimum, the quadratic approximation $f(\mathbf{x}_k^* + \boldsymbol{\delta}) - (f^* - h) \approx \frac{h}{2\sigma_r^2} Y$ with $Y = \sum_{d=1}^{D} \delta_d^2$ is an increasing affine map of Y, under which relative error and kurtosis are invariant. For one coordinate, $E[\delta^2] = \sigma_r^2/3$, $\text{Var}[\delta^2] = 4\sigma_r^4/45$, and $E[(\delta^2 - E[\delta^2])^4] = 16\sigma_r^8/945$, giving single-term excess kurtosis $\frac{16/945}{(4/45)^2} - 3 = \frac{15}{7} - 3 = -\frac{6}{7}$; for a sum of D i.i.d. terms the excess kurtosis scales as $1/D$, hence $-6/(7D)$. The variance of the sample-variance estimator underlying the standard error of $\hat{\mu}$ satisfies $\text{Var}[\hat{\sigma}^2] \approx \sigma^4(\kappa_{\text{excess}} + 2)/M$ to leading order in M; imposing relative standard error $\leq \varepsilon$ with confidence $1-\delta$ under a normal approximation and substituting $\kappa_{\text{excess}} = -6/(7D)$ yields (13). □

We use $\varepsilon = 0.5$ and $\delta = 0.1$ ($z_{0.05} = 1.645$), under which M_{stat} grows only slowly with D (from 18 at $D = 2$ to 22 at $D = 20$) because the perturbed-objective distribution approaches Gaussian as D increases. The *directional coverage bound* $M_{\text{cov}}(D) = 2D$ addresses a complementary failure mode: in high dimensions, a small perturbation sample may under-represent one or two coordinate axes, underestimating the expected fitness of anisotropic peaks that are narrow specifically along those axes. The status of the two bounds differs and should be stated plainly: M_{stat} is a derived statistical guarantee, whereas $M_{\text{cov}} = 2D$ is an explicitly heuristic requirement, namely two independent draws per axis on average, with no worst-case guarantee of covering all axes simultaneously. The two bounds are combined by taking the maximum:

$$M^*(D) = \max\big(M_{\text{stat}}(D),\, M_{\text{cov}}(D)\big). \quad (14)$$

The resulting sample count is $M^*(D) = 18, 19, 20, 21, 40$ at $D = 2, 3, 5, 10, 20$, respectively.

For $D \leq 15$ the statistical bound dominates and $M^* \approx 18$–22, so robustness evaluation costs only 18–22 evaluations per candidate regardless of dimension; at $D = 20$ the coverage bound takes over, doubling the count to 40. These modest values let Phase 2 afford multiple evaluation rounds even for larger candidate pools. An offline audit (released with the code) further confirms that the resulting rankings are insensitive to M: for every $M \geq 10$, the ranking of returned solutions matches a 100,000-sample Monte-Carlo reference exactly on every run.

E. Computational Complexity

Time complexity. Phase 1 performs $O(B_1)$ function evaluations, each followed by $O(k \cdot D)$ distance computations for k-NN rival selection, where $k = O(1)$. Phase 2 evaluates robustness for at most $O(n_{\text{sol}})$ candidates, each requiring $M^*(D) = O(D)$ samples. Overall time complexity is $O(B \cdot D)$, matching standard differential evolution.

Space complexity. The population and archive together store $O((\text{NP} + \text{cap}) \cdot D) = O(N_{\text{est}} \cdot D)$ real vectors.

V. Experimental Results

A. Experimental Setup

1) Comparison Algorithms: We compare TSRDE against eight representative algorithms: DREA [16], SRI [31], PBRO [32], RPSO-IC [33], NCDE [25], LIPS [26], FOA [34], and FastNichingEP (FastNEP) [35]. Four baselines incorporate some form of robustness estimation: DREA integrates expected-fitness evaluation throughout the search, SRI ranks located peaks by a worst-case criterion over sampled perturbation neighborhoods, PBRO augments the fitness with a penalty on perturbation-induced variance, and RPSO-IC applies implicit averaging with conditional acceptance. The remaining four (NCDE, LIPS, FOA, and FastNEP) are conventional MMO algorithms from the crowding, species, and swarm niching lineages that treat all located peaks equally. The set covers the two families relevant to the RMOP rather than ranking state-of-the-art MMO solvers; DREA (2024) is, to our knowledge, the most recent dedicated robust-MMO method. Parameters follow the recommended settings in the respective papers and are listed in Table II; the per-evaluation perturbation-sample counts of the robust baselines are set so that all algorithms exhaust the same evaluation budget, and no baseline-specific tuning was performed on either benchmark.

TABLE II
COMPLETE ALGORITHM PARAMETER SETTINGS. TSRDE'S INTERNAL PARAMETERS ARE ALL DERIVED FROM THE DIMENSION D, THE SEARCH RANGE, AND THE BUDGET B; EXPRESSIONS ARE QUOTED VERBATIM FROM THE REFERENCE IMPLEMENTATION, AND THE OUTER MIN/MAX BOUNDS ARE DEFENSIVE GUARDS THAT REMAIN INACTIVE UNDER THE SETTINGS OF THIS SECTION (NP $\geq$ 80 THROUGHOUT), WITH SECTION IV STATING THE CORRESPONDING ACTIVE FORMS.

Algorithm	Key Parameters	Values
TSRDE	n_{sol} (only user param.)	n_{rob} per problem (1–4; RobBench), 20 (CEC2026)
DREA	robustness evaluation	H=30 samples (original: 100), $\delta = 1\%$ of range, stage-1 fraction 0.15, NP†
SRI	robustness evaluation	H=8, $\delta = 1\%$ of range, stage-1 fraction 0.5, NP†
PBRO	variance penalty	H=8, penalty weight $w_{\max}$=1.0, $\delta = 1\%$ of range, NP†
RPSO-IC	implicit averaging	H=8, $\delta = 1\%$ of range, swarm†
NCDE	population size, F, CR	100, 0.9, 0.1 ($k \in [5, 20]$, restart patience 20)
LIPS	population size, neighbors	100, $n_b \in [2, 5]$
FOA	area limit, lifetime, seeding	100, 10, local/global seeding 2 dims each, step 5% of range
FastNEP	population size, tournament	20, round-robin q=10

†NP/swarm size $= \min(100, \max(20, 5D))$.

2) Benchmarks and Metrics: **RobBench** (Section III): 20 problems across five groups. For each problem, n_{sol} is set equal to the ground-truth number of robust peaks n_{rob}, the ideal setting that directly tests whether Phase 2 selection matches the benchmark's annotation (sensitivity to this choice is examined in Section V-E). The evaluation budget is $20{,}000 \times D$ function evaluations, and performance is measured primarily by RobF1.

CEC2026 MMO Benchmark: the official track of the 2026 IEEE CEC competition on multimodal optimization, comprising 16 problem types across four dimensions ($D \in \{2, 5, 10, 20\}$), with a fixed instance index (pin = 1) per type and dimension. We use the official competition score score $=$ $(\text{RPR} + F_1)/2$ with $n_{\text{sol}} = 20$ and the standard evaluation budget. For this benchmark, TSRDE is evaluated in MMO mode (Phase 2 returns all candidates without robustness-based culling), providing a fair comparison of pure multimodal discovery capability.

3) Protocol: All experiments are conducted on a 96-thread server (754 GB RAM, AMD EPYC). Each algorithm is run independently 10 times per problem instance with different random seeds; seeds are generated deterministically from the run index and algorithm identifier to ensure reproducibility. Statistical significance is assessed via the Wilcoxon signed-rank test on the pooled paired per-run results (200 pairs on RobBench, 640 problem-dimension-run pairs on CEC2026), with Bonferroni correction across the eight baseline comparisons ($\alpha = 0.05$); TSRDE is significantly superior to every baseline on both benchmarks ($p < 10^{-5}$ in all cases), as noted in the captions of Tables III and IV. Raw results, per-problem tables, analysis scripts, and an extended version of this paper are released alongside the source code.

B. Results on RobBench

Table III reports group-average RobF1 on RobBench; per-problem results for all 20 problems are released with the code. TSRDE achieves an overall mean of **0.952**, compared to 0.873 for DREA and 0.827 for SRI, the two strongest baselines; the margin over every baseline is statistically significant overall (Wilcoxon signed-rank, Bonferroni-corrected $\alpha = 0.05$). By mean RobF1, TSRDE is the best-performing algorithm on 19 of the 20 individual problems, the sole exception being RB-A1, where DREA reaches a perfect score.

Several patterns deserve attention.

Groups B and E: consistent perfect scores. TSRDE achieves RobF1 = 1.000 on all four Group B problems and all four Group E problems. Group B spans robustness fractions from 1 robust peak in 4 (RB-B1) to 3 in 4 (RB-B4); Group E is the most structurally deceptive setting, placing the wide-basin peak in a spatially remote corner while packing several narrow peaks near the origin, a gravitational trap for any algorithm that responds primarily to peak density. The conventional MMO baselines score only 0.239–0.352 on Group E (Table III), spreading their recommendations across whichever peaks their niching mechanism converges to, with no robustness criterion. NCDE illustrates a related but distinct failure mode: on RB-E1 all of its returned solutions repeatedly converge to the single robust peak's niche (Fig. 4), yet its precision is penalized

TABLE III
ROBF1 GROUP AVERAGES (±STD ACROSS 4 PROBLEMS PER GROUP) AND OVERALL MEAN ON ROBBENCH. BEST PER ROW IN **BOLD**. PER-PROBLEM RESULTS (20 ROWS) ARE RELEASED WITH THE CODE. TSRDE IS SIGNIFICANTLY BETTER THAN EVERY BASELINE ON THE OVERALL MEAN (WILCOXON SIGNED-RANK, BONFERRONI-CORRECTED α=0.05).

Group	TSRDE	DREA	SRI	NCDE	LIPS	FOA	FastNEP	PBRO	RPSO-IC
A	**0.887±0.121**	0.762±0.196	0.637±0.192	0.475±0.124	0.425±0.082	0.417±0.154	0.467±0.154	0.500±0.000	0.500±0.000
B	**1.000±0.000**	0.946±0.040	0.958±0.028	0.359±0.063	0.317±0.064	0.317±0.064	0.331±0.058	0.583±0.278	0.583±0.278
C	**0.987±0.027**	0.860±0.110	0.854±0.116	0.412±0.144	0.351±0.200	0.385±0.185	0.428±0.164	0.521±0.299	0.521±0.299
D	**0.886±0.114**	0.808±0.138	0.733±0.193	0.314±0.057	0.306±0.055	0.306±0.055	0.322±0.072	0.458±0.073	0.458±0.073
E	**1.000±0.000**	0.988±0.025	0.950±0.100	0.289±0.049	0.239±0.033	0.352±0.097	0.283±0.042	0.750±0.250	0.750±0.250
Overall	**0.952**	0.873	0.827	0.370	0.328	0.355	0.366	0.562	0.562

because it submits far more candidate solutions than there are robust peaks to cover. TSRDE avoids both failure modes: Phase 1's restarts give the search repeated opportunities to locate the isolated peak, and Phase 2's expected-fitness comparison with the spatial-diversity penalty in (12) retains only one solution per basin (Fig. 4).

Group D: the value of expected fitness over variance. Group D problems are anisotropic: the "robust" peaks are wide in most dimensions but noticeably narrower in one or two. This is the failure case for variance-based robustness measures (Section II): variance is dominated by the single narrowest dimension, so a peak that is wide in aggregate can appear fragile. The experimental consequence is qualitative: an earlier variance-based implementation of TSRDE scored only 0.784 on Group D; adopting expected fitness as the ranking criterion raised this to **0.886** (Table III), including RB-D1 improving from 0.750 to a perfect 1.000. The choice of robustness measure is not a secondary implementation detail on anisotropic landscapes.

SRI and DREA as the strongest non-TSRDE algorithms. SRI (0.827 overall) and DREA (0.873 overall) are the only baselines that come close to TSRDE; DREA's perfect score on RB-A1 is the single case where a baseline outperforms TSRDE on an individual problem. Both perform well because they incorporate robustness-aware fitness evaluation, which correlates with ground-truth basin width on isotropic problems. Their limitation emerges on the anisotropic Group D (SRI drops to 0.733 and DREA to 0.808, against TSRDE's 0.886), because a single perturbation-smoothed fitness value cannot cleanly separate a peak that is wide in most dimensions but narrow in one from a peak that is uniformly narrow. At $D = 10$ (RB-A4), SRI's mean RobF1 falls to the level of the random baseline (0.500) while TSRDE maintains 0.700. DREA's remaining deficit is consistent with the budget-allocation hypothesis behind TSRDE's design: robustness evaluation throughout the search leaves less budget for discovering the full set of basins.

C. Independent Validation on CEC2026 (MMO Mode)

This experiment serves a deliberately limited purpose: it validates Phase 1's multimodal discovery capability on a third-party benchmark designed independently of this work by the CEC2026 competition organizers, thereby ruling out benchmark-specific over-fitting as the explanation for the RobBench results (TSRDE was never tuned to its functions; the only parameters adjusted between the two benchmarks are n_{sol} and the MMO/robust mode switch). It is *not* evidence about robustness identification: TSRDE runs in MMO mode here (Phase 2 returns all candidates without robustness-based culling), and the CEC2026 competition score measures conventional peak coverage only. All robustness evidence in this paper is confined to RobBench.

Table IV reports the competition score across dimensions for the CEC2026 MMO benchmark. TSRDE achieves an overall score of **0.550**, compared to 0.302 for the next-best algorithm (FastNEP), a relative improvement of 82%. The advantage is statistically significant at all four dimensions.

TABLE IV
CEC2026 BENCHMARK SCORE (MEAN ± STD, 10 RUNS). BEST PER COLUMN IN **BOLD**. TSRDE IS SIGNIFICANTLY SUPERIOR TO ALL BASELINES AT EVERY DIMENSION (WILCOXON, BONFERRONI, α=0.05).

Algorithm	$D = 2$	$D = 5$	$D = 10$	$D = 20$	Overall
TSRDE	**0.952±0.104**	**0.636±0.226**	**0.353±0.241**	**0.259±0.213**	**0.550**
FastNEP	0.791±0.148	0.291±0.116	0.097±0.074	0.030±0.041	0.302
LIPS	0.318±0.111	0.144±0.062	0.096±0.041	0.050±0.038	0.152
NCDE	0.205±0.089	0.120±0.055	0.091±0.059	0.104±0.071	0.130
FOA	0.337±0.144	0.108±0.061	0.043±0.031	0.021±0.023	0.127
SRI	0.287±0.131	0.049±0.084	0.015±0.033	0.007±0.014	0.089
DREA	0.084±0.057	0.008±0.018	0.001±0.003	0.000±0.000	0.023
RPSO-IC	0.047±0.032	0.009±0.010	0.002±0.004	0.000±0.000	0.014
PBRO	0.029±0.021	0.015±0.012	0.009±0.010	0.004±0.007	0.014

TSRDE's absolute advantage over FastNEP peaks at $D = 5$ (0.636 vs. 0.291) and remains substantial through $D = 10$; its relative advantage instead grows with dimension, reaching an 8.6× ratio at $D = 20$ as FastNEP's score collapses toward zero. NCDE is the only baseline that holds appreciable performance at this dimensionality (0.104): its small-step crowding keeps population elements densely packed in high-dimensional spaces at the cost of slow convergence in low dimensions, whereas TSRDE's dimension-adaptive α reallocates more Phase 2 budget at high dimensions ($\alpha = 0.30$ at $D = 20$) to compensate for the harder sampling problem. The per-problem breakdown (released with the code) shows that TSRDE retains the best $D = 20$ score on 11 of the 16 problems; the five losses, all to NCDE, concentrate on the composition- and transformation-type problems (pid05–07, pid13–14), whose rotated and shifted subcomponents penalize the k-NN rival DE's exploration at that scale. Even so, the absolute performance of all algorithms at $D = 20$ reflects the fundamental difficulty of multimodal optimization at that scale: no algorithm exceeds 0.26, and high-dimensional peak discovery remains an open challenge for the field.

Two further failure modes in the baseline set are worth noting. SRI collapses almost completely at $D \geq 10$ (0.015

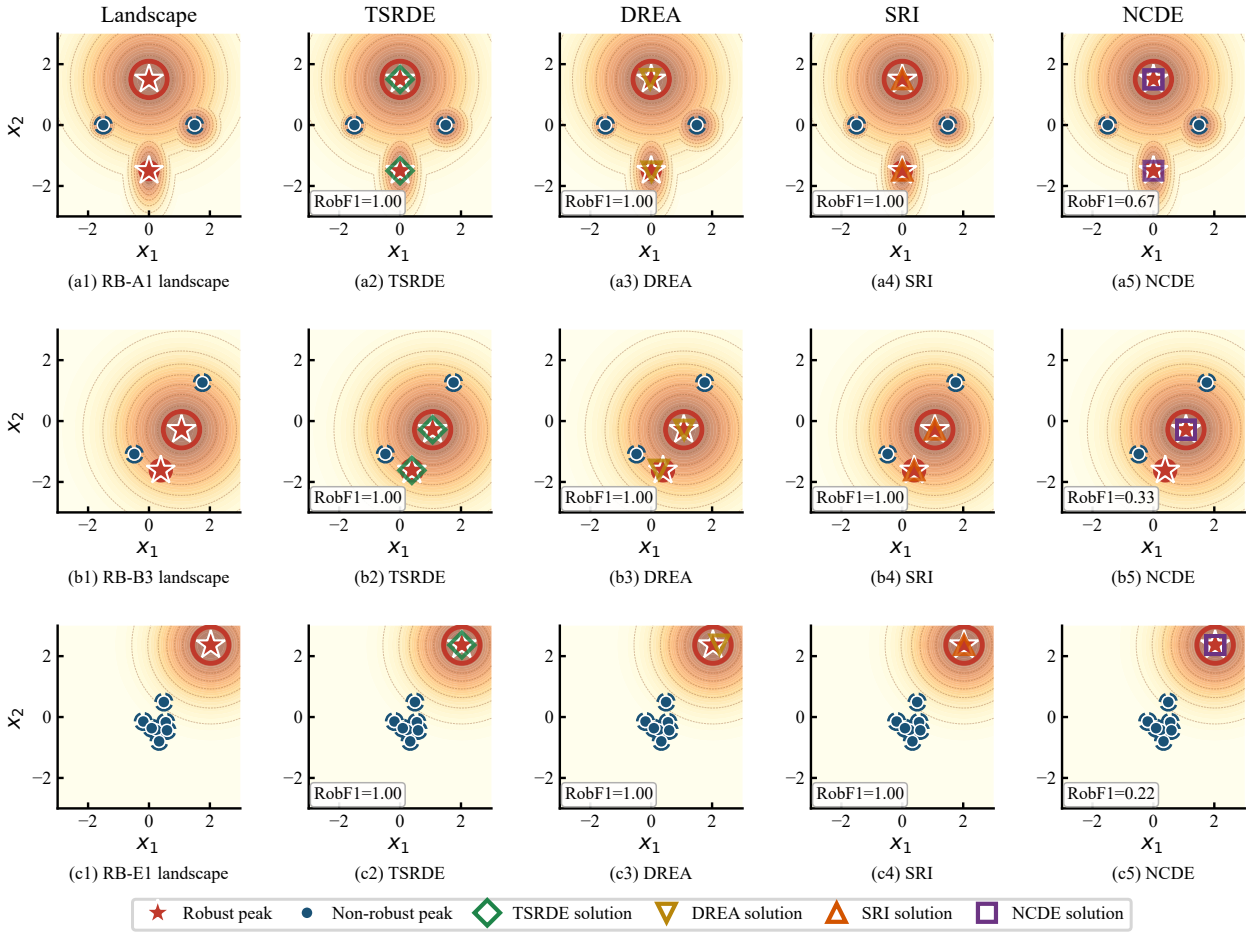


Fig. 4. Solution distributions on three representative RobBench problems (rows) for four algorithms (columns), using the run closest to each algorithm's 10-run median RobF1. (a) RB-A1; (b) RB-B3; (c) RB-E1. Stars denote robust peaks, filled circles non-robust peaks. TSRDE consistently returns one solution per robust peak; DREA and SRI occasionally miss or misassign a peak; NCDE submits redundant solutions converging to the same basin, most visibly in panel (c5).

and 0.007 at $D = 10$ and $D = 20$): its single-peak fitness perturbation scheme neither maintains population diversity across distant basins nor provides reliable directional signal in high dimensions, so the algorithm degenerates to a near-random search. DREA scores only 0.023 overall despite its explicit robustness machinery: continuous robustness evaluation throughout the search leaves insufficient budget for discovering the many peaks that the CEC2026 problems demand. Both patterns are consistent across the two benchmarks and corroborate the design choice to defer robustness assessment to Phase 2.

D. Ablation Study

To isolate the contribution of each component, we evaluate three ablated variants of TSRDE on the full RobBench suite (10 runs per problem, same seeds as the main experiment, identical codebase). **Abl-MMO** disables Phase 2's robustness-aware refinement entirely, returning the n_{sol} archive members with the best nominal fitness after Phase 1; it measures trusting fitness as a proxy for basin width. **Abl-noRestart** keeps Phase 2 intact but disables the stagnation-triggered restart of Section IV-B2; it isolates the value of repeated escapes from converged configurations. **Abl-noElim** replaces the competitive elimination loop (Step 3 of Section IV) with a single uniform-random draw of n_{sol} solutions from the pool surviving the initial robustness-based cull.

Table V reports group-level results.

TABLE V
ABLATION STUDY: MEAN ROBF1 BY PROBLEM GROUP (10 RUNS). RELATIVE CHANGE VS. FULL TSRDE IN PARENTHESES.

Variant	Grp-A	Grp-B	Grp-C	Grp-D	Grp-E	Overall
TSRDE (full)	0.887	1.000	0.987	0.886	1.000	**0.952**
Abl-MMO	0.775	0.988	0.963	0.671	1.000	0.879 (−7.6%)
Abl-noRestart	0.854	1.000	0.977	0.860	1.000	0.938 (−1.4%)
Abl-noElim	0.621	0.721	0.759	0.713	0.721	0.707 (−25.8%)

The most damaging ablation by a wide margin is removing the elimination loop (Abl-noElim, -25.8% overall), and its penalty is broad rather than concentrated (-19.6% to -30.0% across all five groups; Table V). The iterative refine-then-eliminate procedure is thus the primary mechanism by which Phase 2 converts a coarse candidate pool into precise, non-

redundant recommendations: a single random draw, even after the robustness screen, frequently selects multiple solutions from the same basin or keeps a lucky non-robust candidate, errors the multi-round elimination corrects by repeatedly re-refining and re-ranking the shrinking pool.

Removing Phase 2 entirely (Abl-MMO) costs -7.6% overall, concentrated on anisotropic Group D (-24.3%), exactly where robustness and nominal fitness are least correlated; Groups B and E are barely affected because nominal-fitness selection already favors their flatter basins. Disabling the restart (Abl-noRestart) has the smallest effect (-1.4% overall): at RobBench's peak counts ($K \leq 8$) a single Phase 1 population usually populates the archive with all relevant basins, so the restart is a safety margin rather than a coverage essential. The ablations thus reposition the components' importance: robustness-aware selection is essential (the criterion in Abl-MMO and, more sharply, the elimination procedure in Abl-noElim), while the restart strategy is comparatively interchangeable.

E. Parameter Sensitivity: Effect of n_{sol}

Two remarks on the status of the single user parameter, n_{sol}, precede the sensitivity results. First, n_{sol} is a *demand-side* parameter, specifying how many alternatives the decision-maker wants, rather than an algorithm-internal tuning knob; unlike the niche-radius parameters of NCDE and LIPS, which the user cannot derive from the task, it carries operational meaning. Second, the main experiments set $n_{\text{sol}} = n_{\text{rob}}$, the ideal setting that uses ground-truth information unavailable in practice; this choice equally benefits every compared algorithm (each receives the same target count) but should be kept in mind when reading the absolute numbers. The sweep below therefore measures what happens when the ideal setting is not available: we ran TSRDE on all 20 RobBench problems with $n_{\text{sol}} \in \{1, 2, 3, 5, 7, 10, 15\}$ and 10 independent runs per combination (1400 runs total), sweeping below and above the ground-truth n_{rob} of each problem; the corresponding visualizations are released with the code.

The results tell a consistent story: performance is highest at $n_{\text{sol}} = n_{\text{rob}}$ and degrades gracefully, and asymmetrically, in both directions. Under-specification ($n_{\text{sol}} < n_{\text{rob}}$) limits recall by construction: the algorithm is asked for fewer solutions than there are robust peaks, so at $n_{\text{sol}} = 1$ the pooled mean RobF1 drops to 0.695, entirely because problems with $n_{\text{rob}} \geq 2$ automatically cap recall at $1/n_{\text{rob}}$. Over-specification instead costs precision, and gradually rather than catastrophically, because the `diverse_rob_select` procedure actively penalizes redundant basin coverage even when filling surplus slots. Pooling all problems, RobF1 peaks at 0.952 exactly at $\Delta n = n_{\text{sol}} - n_{\text{rob}} = 0$ and falls off in both directions, to 0.834 at $\Delta n = +1$, 0.700 at $\Delta n = -1$, and 0.400 at $\Delta n = -3$, while moderate over-specification ($\Delta n \in \{+1, \ldots, +5\}$) fluctuates in the 0.74–0.83 range rather than declining monotonically; under-specifying n_{sol} is thus the more consequential error. Group-level differences are consistent: Group D is the most forgiving of over-specification, since its anisotropic robust peaks remain the best expected-fitness candidates regardless of surplus slots, whereas Group A is the most sensitive, its closely-spaced isotropic peaks giving the diversity penalty less separation to exploit.

In practice, $n_{\text{sol}} = n_{\text{rob}}$ is the ideal setting when the number of robust peaks is known. When it is unknown (the typical situation), we recommend $n_{\text{sol}} = 3$ as a default: every RobBench problem has $n_{\text{rob}} \leq 4$, so $n_{\text{sol}} = 3$ stays within $\Delta n \leq 2$ everywhere and yields an overall mean RobF1 of 0.827, close to the $n_{\text{sol}} = 2$ pooled optimum of 0.864. Removing the parameter altogether is conceivable, since the Phase 1 archive already contains an estimate of the number and spacing of distinct basins from which n_{sol} could in principle be inferred; we leave such an automatic estimation strategy as future work.

VI. Discussion

The three ablations, read together with the n_{sol} sweep (Section V-E), explain the mechanism behind the full algorithm's behavior. Phase 1 determines *which* basins are available for selection, and its archive is usually complete even without restarts at the peak counts studied (Abl-noRestart); Phase 2 determines *which* of the available basins are returned. Within Phase 2, the expected-fitness criterion distinguishes wide from narrow basins (Abl-MMO, concentrated on anisotropic Group D where fitness and width are least correlated), while the elimination loop converts a *pool* of robust candidates into a *set* of non-redundant recommendations (Abl-noElim, uniformly damaging). The elimination loop's dominance (Table V) reflects the structure of the RMOP itself: with a candidate pool only moderately larger than n_{sol} after quality screening, and robustness estimates that remain noisy at practical sample counts, the binding difficulty is not deciding *whether* a basin is robust but allocating a fixed number of recommendation slots among several candidates, exactly the refinement-and-reranking decision the loop automates.

VII. Conclusion

This paper introduced the robust multimodal optimization problem (RMOP): finding a small set of near-optimal, spatially diverse solutions drawn from the wide-basin optima of an equal-height multi-peaked landscape. To make this problem measurable and solvable, we contributed three coordinated tools. RobBench provides 20 analytically defined test problems whose robustness ground truth is derived directly from the function parameters. RobF1 jointly penalizes redundant recommendations and missed robust peaks, filling a gap left by coverage-only metrics. TSRDE decouples global peak exploration from robustness evaluation through a two-stage design with dimension-adaptive sampling and competitive elimination; all internal parameters are derived from the dimension and search range, leaving n_{sol} as the single user input.

On RobBench, TSRDE significantly outperformed the eight tested multimodal and robust baselines in RobF1 (mean 0.952). The independent CEC2026 evaluation, in which TSRDE runs in MMO mode, indicated that this robustness

capability does not come at the cost of conventional peak-finding performance. The robustness conclusions are established on equal-height Gaussian-mixture landscapes with moderate anisotropy; their extension to heterogeneous-height or non-Gaussian landscapes remains to be demonstrated. Natural next steps include worst-case and risk-aware robustness criteria beyond the expected fitness, a multi-objective formulation of the RMOP with RobBench-style ground truth, and automatic estimation of n_{sol} from Phase 1 archive statistics. We hope that RobBench and RobF1 provide the community with principled tools for measuring and advancing progress on robust solution discovery.

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Appendix for: Two-Stage Robust Differential Evolution for Multimodal Optimization

This document is the appendix of the extended version of the paper, organized into eight sections. Section S-I derives the statistical and directional sample-count bounds $M_{\text{stat}}(D)$ and $M_{\text{cov}}(D)$. Section S-II gives the full 20-problem RobBench specification. Section S-III lists the complete parameter settings for all nine compared algorithms, together with the JADE parameter-adaptation equations summarized in the main text. Sections S-IV and S-V report per-problem RobF1 and CEC2026 competition scores, respectively; Section S-IV additionally provides a group-level visualization of the RobBench results (Fig. S-2) and of the ablation study (Fig. S-3), and Section S-V adds representative CEC2026 solution distributions (Fig. S-4). Section S-VI tabulates the dimension-adaptive sample count $M^*(D)$ across all tested dimensions (Table S-V, Fig. S-5). Section S-VII presents the complete n_{sol} sensitivity figures and summary table. Section S-VIII elaborates the RobF1 metric design rationale, a near-optimality audit of all returned solutions, and the per-problem verification that the expected-fitness criterion reproduces RobBench's robustness ordering. In the 12-page submission version of the paper, these materials are cited collectively as released-with-the-code resources; the per-problem tables are additionally provided in Markdown form in the same repository.

S-I. Derivation of Statistical and Coverage Sample-Count Bounds

This section expands the statistical precision bound of the main paper's Section "Dimension-Adaptive Sampling for Robustness Estimation," detailing the intermediate moment calculations underlying Proposition 2 there.

A. Setup

Near the optimum of an isotropic peak with height h and width σ_k (i.e., $s_{k,d} = \sigma_k$ for all d), the quadratic approximation of the perturbed objective is

$$f(\mathbf{x}_k^* + \boldsymbol{\delta}) - \big(f^* - h\big) \approx \frac{h}{2\sigma_k^2} \sum_{d=1}^{D} \delta_d^2, \tag{S1}$$

where $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_D)$ with $\delta_d \sim U(-\sigma_r, \sigma_r)$ independently. Define $Y = \sum_{d=1}^{D} \delta_d^2$, so that the perturbed-objective excess is $\frac{h}{2\sigma_k^2} Y$ up to an additive constant; all kurtosis and relative-error calculations below are invariant to this positive affine transform and therefore apply directly to $f(\mathbf{x}_k^* + \boldsymbol{\delta})$.

B. Moments of $Y = \sum_d \delta_d^2$

For a single coordinate $\delta \sim U(-\sigma_r, \sigma_r)$, the even moments of δ^2 are

$$E[\delta^2] = \frac{\sigma_r^2}{3}, \qquad E[\delta^4] = \frac{\sigma_r^4}{5}. \tag{S2}$$

Hence, using also $E[\delta^6] = \sigma_r^6/7$ and $E[\delta^8] = \sigma_r^8/9$,

$$\mathrm{Var}[\delta^2] = E[\delta^4] - E[\delta^2]^2 = \frac{\sigma_r^4}{5} - \frac{\sigma_r^4}{9} = \frac{4\sigma_r^4}{45}, \tag{S3}$$

$$E\big[(\delta^2 - E[\delta^2])^4\big] = \frac{16\sigma_r^8}{945}. \tag{S4}$$

Since the D coordinates are i.i.d., $Y = \sum_d \delta_d^2$ has

$$E[Y] = D \cdot \frac{\sigma_r^2}{3}, \qquad \mathrm{Var}[Y] = D \cdot \frac{4\sigma_r^4}{45}. \tag{S5}$$

C. Excess Kurtosis of Y

For a sum of D i.i.d. terms, the excess kurtosis satisfies the standard scaling law

$$\kappa_{\text{excess}}[Y] = \frac{\kappa_4[\delta^2]}{D}, \tag{S6}$$

where $\kappa_4[\delta^2]$ is the excess kurtosis of a single term δ^2. Using the central moments above,

$$\kappa_4[\delta^2] = \frac{E[(\delta^2 - E[\delta^2])^4]}{\mathrm{Var}[\delta^2]^2} - 3 = \frac{16\sigma_r^8/945}{(4\sigma_r^4/45)^2} - 3 = \frac{16/945}{16/2025} - 3. \tag{S7}$$

Simplifying $\frac{16/945}{16/2025} = \frac{2025}{945} = \frac{15}{7}$, so $\kappa_4[\delta^2] = \frac{15}{7} - 3 = -\frac{6}{7}$ (the squared-uniform distribution is platykurtic). Substituting into (S6):

$$\kappa_{\text{excess}}[Y] = -\frac{6}{7D}, \tag{S8}$$

matching the value quoted in Section IV-D of the main paper. Because the perturbed-objective excess is an increasing affine function of Y (larger perturbation displacement always increases f near a minimum), this negative excess kurtosis carries over directly to $f(\mathbf{x}_k^* + \boldsymbol{\delta})$: the perturbed-objective distribution is slightly flatter than Gaussian at low D, with the deviation vanishing as $D \to \infty$ by the central limit theorem.

D. From Kurtosis to Sample Count

For a sample mean $\hat{Y} = \frac{1}{M}\sum_{i=1}^{M} Y_i$ of M i.i.d. draws, the variance of the *sample variance* estimator that governs the standard error of $\hat{Y}$ satisfies, to leading order in M,

$$\mathrm{Var}[\hat{\sigma}^2] \approx \frac{\sigma^4}{M}\Big(\kappa_{\text{excess}} + 2\Big), \tag{S9}$$

a standard result for the sampling variance of the variance estimator (e.g., via the delta method applied to the fourth central moment). Requiring the relative standard error of $\hat{Y}$ (equivalently, of $\hat{\mu}(\mathbf{x})$ under the affine map) to satisfy RSE $\leq \varepsilon$ with confidence level $1-\delta$ under a normal approximation gives the sample-size requirement

$$M_{\text{stat}} = \left\lceil \frac{z_{\delta/2}^2 \cdot (\kappa_{\text{excess}} + 2)}{\varepsilon^2} \right\rceil, \tag{S10}$$

which, substituting $\kappa_{\text{excess}} = -6/(7D)$ and the default $\delta = 0.1$ ($z_{0.05} = 1.645$, $z_{0.05}^2 = 2.706$), $\varepsilon = 0.5$, recovers the main paper's M_{stat} formula:

$$M_{\text{stat}}(D) = \left\lceil \frac{2.706\,\big(2 - 6/(7D)\big)}{0.25} \right\rceil. \tag{S11}$$

Table S-V (Table S-V below) tabulates $M_{\text{stat}}(D)$, $M_{\text{cov}}(D) = 2D$, and $M^*(D) = \max(M_{\text{stat}}, M_{\text{cov}})$ across the full tested dimension range.

E. Directional Coverage Bound

The statistical bound $M_{\text{stat}}(D)$ above guarantees precision of the sample mean *on average* over the D-dimensional perturbation ball, but it does not by itself guarantee that every coordinate direction is represented in a finite sample of size M. This matters specifically for anisotropic peaks (Section "RobBench," Group D of the main paper): if a peak is narrow along one dimension $d^\dagger$ but wide along all others, a sample $\{\boldsymbol{\delta}_i\}_{i=1}^M$ that happens to place little perturbation mass along $d^\dagger$ will underestimate the true expected fitness, making the peak appear more robust than it is. Requiring at least two independent draws per coordinate axis on average,

$$M_{\text{cov}}(D) = 2D, \tag{S12}$$

provides a simple, distribution-free safeguard against this systematic under-sampling. This bound is intentionally weak: it does not guarantee simultaneous coverage of all D axes in the worst case, only that the *expected* number of samples per axis is at least two. Combined with the random (not grid-based) perturbation sampling used in Phase 2, it is sufficient to prevent the narrow-axis blind spot from dominating the robustness estimate. Taking $M^*(D) = \max(M_{\text{stat}}(D), M_{\text{cov}}(D))$ ensures neither failure mode (imprecise mean, under-sampled axis) is left unaddressed.

S-II. RobBench Problem Specifications

Table S-I lists all 20 RobBench problems with their complete parameters, computed directly from the benchmark source (peak centers, per-dimension widths, and the analytically derived ground-truth robust set). $\sigma_{\min}/\sigma_{\max}$ are the minimum/maximum mean peak width across the K peaks in each problem; "Random baseline" is n_{rob}/K, the expected RobF1 recall achievable by an algorithm that selects peaks uniformly at random.

Note: Group D's $\sigma_{\max} = 1.20$ (rather than 1.50 as in the isotropic groups A/B/C) reflects that each Group D problem contains one isotropic wide-basin peak at $\sigma = 1.20$ and one anisotropic robust peak whose per-dimension widths are mixed: wide in most dimensions but narrow (0.30) in one or two. For example, RB-D1's anisotropic robust peak has per-dimension widths $[0.80, 0.80, 0.80, 0.80, 0.30]$, giving $\sigma_k = 0.70$ (as stated in the main text). Both this anisotropic peak ($\sigma_k = 0.70$) and the isotropic peak ($\sigma_k = 1.20$) are robust; the two remaining peaks are non-robust with $\sigma_k = 0.30$.

TABLE S-I
COMPLETE ROBBENCH PROBLEM SPECIFICATION (20 PROBLEMS).

Group	Problem	D	K	n_{rob}	$\sigma_{\min}$	$\sigma_{\max}$	Baseline
A	RB-A1	2	4	2	0.30	1.50	0.500
	RB-A2	3	4	2	0.30	1.50	0.500
	RB-A3	5	4	2	0.30	1.50	0.500
	RB-A4	10	4	2	0.30	1.50	0.500
B	RB-B1	5	4	1	0.30	1.50	0.250
	RB-B2	5	6	2	0.30	1.50	0.333
	RB-B3	5	4	2	0.30	1.50	0.500
	RB-B4	5	4	3	0.30	1.50	0.750
C	RB-C1	5	2	1	0.35	1.00	0.500
	RB-C2	5	4	2	0.30	1.50	0.500
	RB-C3	5	6	3	0.30	1.50	0.500
	RB-C4	5	8	4	0.30	1.50	0.500
D	RB-D1	5	4	2	0.30	1.20	0.500
	RB-D2	10	4	2	0.30	1.20	0.500
	RB-D3	5	4	2	0.30	1.20	0.500
	RB-D4	10	6	3	0.30	1.20	0.500
E	RB-E1	5	8	1	0.30	1.20	0.125
	RB-E2	10	8	2	0.30	1.00	0.250
	RB-E3	5	6	1	0.32	1.50	0.167
	RB-E4	10	6	2	0.30	1.00	0.333

Fig. S-1 visualizes all 20 problems as 2-D contour plots, complementing the representative subset shown in Fig. 2 of the main paper.

S-III. Algorithm Parameter Settings

Table S-II lists the complete parameter settings for all nine compared algorithms, expanding the main paper's abbreviated description. The four conventional MMO baselines (NCDE, LIPS, FOA, FastNEP) use their originally published default hyperparameters. The four robustness-oriented baselines (DREA, SRI, PBRO, RPSO-IC) are faithful re-implementations of the original algorithmic schemes, with the per-evaluation perturbation-sample count H set to 8 (30 for DREA) so that all algorithms exhaust the same evaluation budget; their perturbation radius is fixed at 1% of the search range. No baseline-specific tuning was performed on either RobBench or CEC2026.

JADE parameter-adaptation equations. The mutation scale factor F and crossover rate CR of Phase 1 are adapted following JADE. After each generation, the F and CR values of trials that successfully replaced their rivals are recorded, and the means are updated as

$$\mu_F \leftarrow (1-c)\,\mu_F + c \cdot \frac{\sum_i F_i^2}{\sum_i F_i}, \tag{S13}$$

$$\mu_{CR} \leftarrow (1-c)\,\mu_{CR} + c \cdot \text{mean}(\{\text{CR}_i\}), \tag{S14}$$

with learning rate $c = 0.1$; μ_F and μ_{CR} are initialized at 0.5 and 0.9. Individual F values are then sampled from a Cauchy distribution centered at μ_F, and CR values from a normal distribution centered at μ_{CR}, both truncated to valid ranges.

S-IV. Per-Problem RobF1 Results on RobBench

Table S-III reports per-problem RobF1 (mean ± std, 10 runs) for all nine algorithms across the 20 RobBench problems. Group averages and the overall mean are summarized in Table III of the main paper. Fig. S-2 visualizes these group-level averages: TSRDE leads every group, with the largest margin over the two closest baselines, DREA and SRI, on the anisotropic Group D. Fig. S-3 additionally visualizes the ablation study of the main paper (Section V-D) at the group level: the four variants (full TSRDE and the three ablations) share the same codebase and seeds, and the per-group bars make the uniform damage of Abl-noElim and the Group D concentration of Abl-MMO directly visible.

S-V. Per-Problem CEC2026 Competition Scores

Table S-IV reports per-problem competition scores (mean ± std, 10 runs, pin = 1) for all nine algorithms across the 16 CEC2026 problem types at four dimensions. Dimension-averaged results are reproduced in the last four rows. Bold entries mark the best-performing algorithm per row.

Three patterns are consistent across problems. **First**, TSRDE leads on the majority of problems at every dimension: all 16 problems at $D = 2$ and $D = 5$, 12 of 16 at $D = 10$, and 11 of 16 at $D = 20$. The exceptions at higher dimensions are concentrated in pid05–07 and pid13–14, where NCDE's small-step crowding mechanism maintains sufficient population density to outperform TSRDE's two-phase strategy under the tighter budget. **Second**, FastNEP is consistently the second-strongest algorithm at $D = 2$, often achieving scores above 0.75, but degrades sharply at $D \geq 5$, confirming the dimension sensitivity described in the main text. **Third**, DREA, PBRO, and RPSO-IC produce near-zero scores at $D \geq 10$ across virtually all problems, reflecting algorithm designs that do not scale to higher dimensions.

Fig. S-4 complements the per-problem scores with representative solution distributions at $D = 2$: on the easy pid01, all four algorithms place solutions near most optima, though DREA's coverage is already visibly scattered; on the medium and hard pid07 and pid15, only TSRDE's solution cloud remains concentrated on the true peaks, while FastNEP and NCDE show progressively sparser coverage and DREA largely misses the optima entirely, consistent with its near-zero scores in Table S-IV.

S-VI. Dimension-Adaptive Sample Count $M^*(D)$

Table S-V tabulates the dimension-adaptive sample count across all five tested dimensions, showing the crossover from the statistical bound M_{stat} to the coverage bound $M_{\text{cov}} = 2D$ at $D = 20$.

TABLE S-V
Dimension-adaptive sample count $M^*(D)$, all tested dimensions.

D	M_{stat}	$M_{\text{cov}} = 2D$	$M^*(D)$	Dominant bound
2	18	4	18	Statistical
3	19	6	19	Statistical
5	20	10	20	Statistical
10	21	20	21	Statistical
20	22	40	40	Coverage

Both $M_{\text{stat}}(D)$ and $M_{\text{cov}}(D) = 2D$ are monotone increasing in D, but M_{stat} grows only logarithmically slowly (from 18 to 22 across the entire tested range) because the excess-kurtosis correction term $6/(7D)$ in (S10) shrinks toward zero as D grows, driving the perturbed-objective distribution toward Gaussian. M_{cov}, by contrast, grows linearly, and overtakes M_{stat} between $D = 10$ and $D = 20$. Fig. S-5 visualizes both bounds and their maximum over $D \in [1, 25]$: $M^*(D)$ remains nearly flat through $D \approx 13$ and then increases linearly, balancing statistical reliability against directional coverage. This crossover is why Phase 2's per-candidate robustness-evaluation cost roughly doubles from $D = 10$ ($M^* = 21$) to $D = 20$ ($M^* = 40$), motivating the dimension-adaptive $\alpha(D)$ budget schedule to reallocate more budget toward Phase 2 at $D = 20$ (main paper, Section IV; Table S-II).

Offline sample-count sensitivity audit. To complement the derivation, we audited how sensitive the expected-fitness *ranking* of TSRDE's returned solutions is to the sample count. For every RobBench run, the returned solutions were re-evaluated offline under the same estimator as in the main paper (uniform perturbation of radius $\sigma_r = 0.1R_s$, clipped to bounds) with $M \in \{5, 10, 20, 40\}$, and compared against a 100,000-sample Monte-Carlo reference. For every $M \geq 10$, Kendall's τ between the M-sample ranking and the reference equals 1.000 on every run (mean absolute estimation error 0.006 at M=10, 0.003 at M=40); at $M = 5$ the mean τ drops to 0.988 with occasional rank inversions (minimum 0.33). Problems returning a single solution are excluded, as their ranking is undefined. Since $M^*(D) \in [18, 40]$ over the whole suite, the chosen sample count lies well inside the regime where the ranking is invariant, and only aggressively small M introduces noise. The scope of this audit should be noted: it re-evaluates final solutions offline and therefore bounds the sensitivity of the estimation stage; it does not measure the effect of M on the search trajectory itself, which would require instrumented re-runs.

S-VII. n_{SOL} Parameter Sensitivity Analysis

Figs. S-6 and S-7 provide the complete visualizations underlying the main paper's Section "Parameter Sensitivity: Effect of n_{sol}," based on the full sweep of $n_{\text{sol}} \in \{1, 2, 3, 5, 7, 10, 15\}$ across all 20 RobBench problems (10 runs each, 1400 runs total).

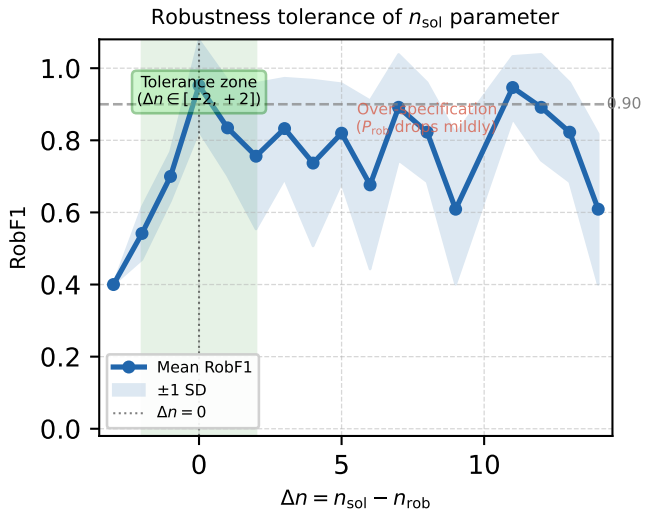


Fig. S-7. RobF1 as a function of $\Delta n = n_{\text{sol}} - n_{\text{rob}}$, pooled across all 20 problems and all seven sweep values. The shaded green band marks $\Delta n \in [-2, +2]$; the dashed line marks RobF1$= 0.90$, reached only at $\Delta n = 0$.

Table S-VI summarizes the practical trade-off: overall mean RobF1 (pooled across all 20 problems and 10 runs) as a function of the fixed n_{sol} value used, independent of each problem's true n_{rob}.

TABLE S-VI
OVERALL MEAN ROBF1 BY FIXED n_{SOL} SETTING (POOLED ACROSS 20 PROBLEMS, 10 RUNS EACH).

n_{sol}	Overall mean RobF1
1	0.695
2	0.864
3	0.827
5	0.799
7	0.795
10	0.796
15	0.796

The overall optimum is $n_{\text{sol}} = 2$ (0.864), reflecting that most RobBench problems have $n_{\text{rob}} = 2$ (12 of 20; Table S-I). $n_{\text{sol}} = 3$ trades a small amount of pooled performance (0.827) for robustness against the common case $n_{\text{rob}} \in \{1, 3, 4\}$, which $n_{\text{sol}} = 2$ would under- or over-specify. Performance is essentially flat for $n_{\text{sol}} \geq 7$ (0.795–0.796), indicating that once enough surplus slots are provided, additional over-specification has negligible further cost.

S-VIII. RobF1 Metric Design Rationale

This section elaborates the metric design choices behind RobF1 (main paper, Section III-D); the three scoring scenarios are illustrated in Fig. 3 of the main paper.

A. Why n_{in} rather than n_{sol} in the precision denominator?

Out-of-niche solutions indicate incomplete convergence rather than erroneous recommendations. Including them in the denominator would double-penalize algorithms that struggle to converge: once for failing to reach any peak, and again for a deflated precision score. By counting only in-niche solutions, P_{rob} assesses the *quality* of converged solutions independently of the *coverage* achieved.

As a concrete illustration, consider an algorithm returning $n_{\text{sol}} = 4$ solutions, three of which lie outside any peak's niche. The single in-niche solution covers a robust peak, so $n_{\text{in}} = 1$, $n_{\text{cov}} = 1$, $P_{\text{rob}} = 1.0$, and $R_{\text{rob}} = 1/n_{\text{rob}}$. Using $n_{\text{sol}} = 4$ in the denominator would give $P_{\text{rob}} = 0.25$, artificially deflating the score for what is, in the converged subset, a perfect precision result.

B. Do solutions converged to non-robust peaks affect P_{rob}?

Yes. A solution within the niche of a non-robust peak increments n_{in} (the denominator) but not n_{cov} (the numerator), thus diluting precision. This correctly penalizes algorithms that recommend narrow peaks, the primary failure mode RobF1 is designed to distinguish. For example, if $n_{\text{rob}} = 2$ but all n_{sol} solutions converge to a single non-robust peak, then $n_{\text{in}} > 0$ but $n_{\text{cov}} = 0$, giving $P_{\text{rob}} = 0$ and RobF1 $= 0$.

C. Redundancy penalty

When multiple solutions cover the same robust peak, n_{in} increments multiple times but n_{cov} increments only once, reducing P_{rob}. This penalizes over-concentration on a single robust basin at the expense of missing others. For example, if $n_{\text{rob}} = 2$ and both returned solutions cluster in the niche of the first robust peak (missing the second), then $n_{\text{in}} = 2$, $n_{\text{cov}} = 1$, $P_{\text{rob}} = 0.5$, $R_{\text{rob}} = 0.5$, and RobF1 $= 0.5$, the redundancy scenario illustrated in Fig. 3(b) of the main paper.

D. Ordering consistency among robustness summaries

Section II-A of the main paper notes that the expected-fitness term $\sum_d s_{k,d}^{-2}$, the harmonic mean $D/\sum_d s_{k,d}^{-1}$, and the arithmetic mean σ_k used as RobBench's ground truth are not order-equivalent in general, and that RobBench's anisotropy is deliberately moderate. We verified this per instance: for every one of the 20 problems, the three criteria were computed from the benchmark's function parameters and compared. On all 20 problems, each pair of criteria induces the identical full ranking of the K peaks (pairwise Kendall's $\tau = 1.000$) and the identical top-n_{rob} set, so the robust/non-robust boundary is the same under all three. The tightest case is RB-A1, where the boundary margin under the expected-fitness criterion, i.e., the relative difference between the $\sum_d s_{k,d}^{-2}$ values of the least-robust robust peak and the most-robust non-robust peak, is 2.3%; on every other problem the margin exceeds 29%. Only the four Group D problems contain anisotropic peaks (one per problem); on isotropic problems the three summaries are monotone transformations of one another and agree trivially. The verification script and its JSON output are released with the code.

E. Near-optimality audit of returned solutions

RobF1 verifies the coverage conditions of the RMOP (conditions 2 and 3 in the main paper) but not near-optimality (condition 1), because niche membership is a spatial criterion. To measure whether returned solutions nevertheless satisfy

condition (1), we audited every solution returned in every RobBench run of all nine algorithms, computing the nominal excess $f(x) - f^*$ with $f^* = -10$.

Three observations follow from Table S-VII. First, all 385 solutions returned by TSRDE satisfy $f(x) - f^* < 10^{-13}$ (maximum observed excess 4.6×10^{-14}) and lie within their niche radius, so for the proposed algorithm RobF1 and near-optimality coincide in practice; the quality screen and local refinement of Phase 2 enforce condition (1) by construction. Second, every solution of every algorithm lies inside some niche (in-niche rate 1.000), yet several baselines return solutions far from f^* (median excess 0.23 for DREA, 1.27 for FastNEP): niche membership demonstrably does not imply near-optimality, which is precisely why condition (1) must be audited separately rather than inferred from RobF1. Third, the audit separates algorithms in a way RobF1 alone does not; for example, DREA's high RobF1 (0.873) is obtained with mostly non-near-optimal solutions, because its robustness-adjusted outputs sit off the peak summits. Solution counts differ by design: conventional MMO algorithms return five solutions per problem, robustness-oriented baselines return n_{rob}, and TSRDE returns 385 rather than 410 because the final deduplication merges near-duplicates on the high-dimensional multi-peak problems (RB-A4, RB-C4, RB-D2, RB-D4) instead of padding the output with redundant solutions. The audit script and its JSON output are released with the code.

TABLE S-II
COMPLETE ALGORITHM PARAMETER SETTINGS. TSRDE'S INTERNAL PARAMETERS ARE ALL DERIVED FROM THE DIMENSION D, THE SEARCH RANGE, AND THE BUDGET B.

Algorithm	Key Parameters	Values
TSRDE	n_{sol} (only user param.)	n_{rob} per problem (1–4; RobBench), 20 (CEC2026)
DREA	robustness evaluation	$H = 30$ samples (original: 100; reduced for the shared FE budget), $\delta = 1\%$ of range, stage-1 fraction 0.15, NP†
SRI	robustness evaluation	$H = 8$, $\delta = 1\%$ of range, stage-1 fraction 0.5, NP†
PBRO	variance penalty	$H = 8$, penalty weight $w_{\max} = 1.0$, $\delta = 1\%$ of range, NP†
RPSO-IC	implicit averaging	$H = 8$, $\delta = 1\%$ of range, swarm†
NCDE	population size, F, CR	100, 0.9, 0.1 ($k \in [5, 20]$, restart patience 20)
LIPS	population size, neighbors	100, $n_b \in [2, 5]$
FOA	area limit, lifetime, seeding	100, 10, local/global seeding 2 dims each, step 5% of range
FastNEP	population size, tournament	20, round-robin $q = 10$

†NP/swarm size $= \min(100, \max(20, 5D))$.

TSRDE internal parameters (all derived from the dimension D, the search range, and the budget B):

N_{est}, R_s	$\max(20, 5D)$; $(\mathbf{u}_1 - \mathbf{l}_1)/(5\, N_{\text{est}}^{1/D})$
population size NP	$\min(\max(4N_{\text{est}},\ 15D),\ \lfloor B/10 \rfloor)$
archive	capacity $\max(6N_{\text{est}}, 80)$; dedup radius R_s ($R_s/4$ in MMO mode)
Phase-1 share α	robust mode: 0.65; MMO mode: 0.65 ($D<10$) / 0.525 ($10 \le D < 20$) / 0.30 ($D \ge 20$)
restart	patience $s^* = \max(5, \min(\lfloor B_1/(12\,\text{NP}) \rfloor, \text{NP}))$ generations; at most 5 restarts; only in the first 90% of B_1
rival crowding	$k_{\text{lcf}} = \min(\text{NP}-1, \max(3, \min(20, 3D)))$; $k_{\text{rival}} = \max(3, \min(10, \lfloor \text{NP}/N_{\text{est}} \rfloor))$, decaying linearly from 4 to 2 over Phase 1
Phase-2 screening	band $\max(0.05(f_{P_{90}} - f_{\text{best}}),\ 0.5(f_{P_{50}} - f_{\text{best}}))$; $K = \max(5n_{\text{sol}}, 15)$; $n_{\text{pre}} = \max(3n_{\text{sol}}, n_{\text{sol}}+2)$
elimination	remove $\lceil 0.33\lvert C \rvert \rceil$ (at least 1) per round; round budget 35% of remaining; local search at least $3D$ per candidate
robustness estimate	$\sigma_r = 0.1R_s$; $M^*(D)$ (Table S-V); local search: Nelder–Mead ($D \le 5$), L-BFGS-B ($D > 5$)
JADE	$c = 0.1$; μ_F, μ_{CR} initialized at $0.5, 0.9$

Expressions are quoted verbatim from the reference implementation. The outer min/max bounds are defensive guards that remain inactive under the experimental settings of the main paper (e.g., NP ≥ 80 throughout, so the NP-1 bound on k_{lcf} never binds); the main text states the corresponding active forms.

TABLE S-III
ROBF1 (MEAN ± STD, 10 RUNS) ON ALL 20 ROBBENCH PROBLEMS. BEST RESULT PER ROW IN **BOLD**. † = TSRDE SIGNIFICANTLY BETTER OVERALL (WILCOXON, BONFERRONI, α=0.05).

Problem	TSRDE	DREA†	SRI†	NCDE†	LIPS†	FOA†	FastNEP†	PBRO†	RPSO-IC†
Group A: Dimensionality Scaling									
RB-A1	0.850±0.229	**1.000±0.000**	0.700±0.458	0.600±0.133	0.533±0.163	0.667±0.000	0.667±0.000	0.500±0.000	0.500±0.000
RB-A2	**1.000±0.000**	0.500±0.000	0.400±0.200	0.633±0.100	0.500±0.167	0.333±0.000	0.533±0.163	0.500±0.000	0.500±0.000
RB-A3	**1.000±0.000**	0.900±0.200	0.950±0.150	0.333±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
RB-A4	**0.700±0.100**	0.650±0.229	0.500±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
Group B: Robustness Fraction									
RB-B1	**1.000±0.000**	**1.000±0.000**	**1.000±0.000**	0.400±0.000	0.400±0.000	0.400±0.000	0.400±0.000	**1.000±0.000**	**1.000±0.000**
RB-B2	**1.000±0.000**	0.950±0.150	0.950±0.150	0.275±0.075	0.250±0.000	0.250±0.000	0.275±0.075	0.500±0.000	0.500±0.000
RB-B3	**1.000±0.000**	0.900±0.200	0.950±0.150	0.333±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
RB-B4	**1.000±0.000**	0.933±0.133	0.933±0.133	0.429±0.143	0.286±0.000	0.286±0.000	0.314±0.086	0.333±0.000	0.333±0.000
Group C: Peak Multiplicity									
RB-C1	**1.000±0.000**	**1.000±0.000**	**1.000±0.000**	0.667±0.000	0.667±0.000	0.667±0.000	0.667±0.000	**1.000±0.000**	**1.000±0.000**
RB-C2	**1.000±0.000**	0.750±0.250	0.850±0.229	0.333±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
RB-C3	**1.000±0.000**	0.967±0.100	0.867±0.163	0.333±0.111	0.222±0.000	0.222±0.000	0.444±0.000	0.333±0.000	0.333±0.000
RB-C4	**0.946±0.086**	0.725±0.135	0.700±0.150	0.317±0.050	0.183±0.050	0.317±0.050	0.267±0.082	0.250±0.000	0.250±0.000
Group D: Anisotropic Geometry									
RB-D1	**1.000±0.000**	0.950±0.150	**1.000±0.000**	0.367±0.100	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
RB-D2	**0.767±0.153**	0.700±0.245	0.750±0.250	0.333±0.000	0.333±0.000	0.333±0.000	0.333±0.000	0.500±0.000	0.500±0.000
RB-D3	**1.000±0.000**	0.950±0.150	0.650±0.229	0.333±0.000	0.333±0.000	0.333±0.000	0.400±0.133	0.500±0.000	0.500±0.000
RB-D4	**0.777±0.196**	0.633±0.180	0.533±0.163	0.222±0.000	0.222±0.000	0.222±0.000	0.222±0.000	0.333±0.000	0.333±0.000
Group E: Spatial Trap									
RB-E1	**1.000±0.000**	**1.000±0.000**	0.800±0.400	0.222±0.000	0.222±0.000	0.222±0.000	0.222±0.000	**1.000±0.000**	**1.000±0.000**
RB-E2	**1.000±0.000**	0.950±0.150	**1.000±0.000**	0.300±0.100	0.200±0.000	0.400±0.000	0.300±0.100	0.500±0.000	0.500±0.000
RB-E3	**1.000±0.000**	**1.000±0.000**	**1.000±0.000**	0.286±0.000	0.286±0.000	0.286±0.000	0.286±0.000	**1.000±0.000**	**1.000±0.000**
RB-E4	**1.000±0.000**	**1.000±0.000**	**1.000±0.000**	0.350±0.122	0.250±0.000	0.500±0.000	0.325±0.115	0.500±0.000	0.500±0.000

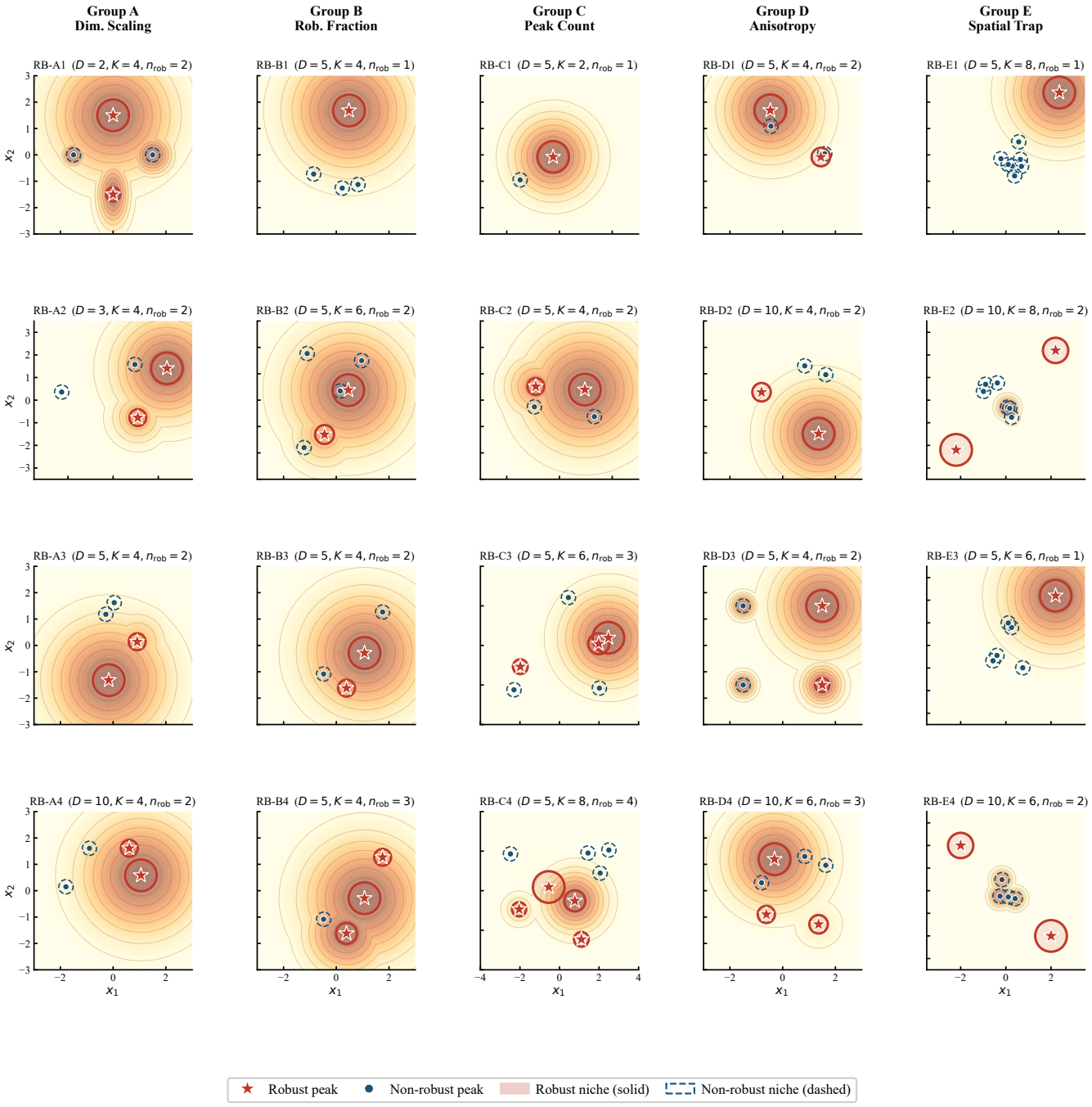


Fig. S-1. All 20 RobBench problem landscapes (5 groups × 4 problems, 5-row × 4-column grid). Each subplot shows a 2-D contour of the objective function evaluated over the first two dimensions (all remaining dimensions fixed at zero). Gold stars (★) and blue dashed circles mark robust peaks and their niche radii; red filled circles and red dotted circles mark non-robust peaks and their niche radii. Problem labels show dimension D, total peak count K, and the number of robust peaks n_{rob}. Note that for $D > 2$ the contour is a 2-D slice: peaks whose centres lie far from zero in the omitted dimensions appear shallower than their true height, but all peak *locations* in the displayed dimensions are correctly shown.

TABLE S-IV
PER-PROBLEM CEC2026 COMPETITION SCORE (MEAN ± STD, 10 RUNS, PIN = 1). BEST RESULT PER ROW IN **BOLD**.

Problem	TSRDE	DREA	SRI	NCDE	LIPS	FOA	FastNEP	PBRO	RPSO-IC
$D = 2$									
pid01	**1.000±0.000**	0.055±0.020	0.216±0.019	0.228±0.050	0.290±0.066	0.301±0.023	0.782±0.038	0.025±0.009	0.035±0.012
pid02	**1.000±0.000**	0.103±0.018	0.247±0.011	0.185±0.072	0.285±0.063	0.273±0.020	0.802±0.036	0.033±0.010	0.080±0.006
pid03	**1.000±0.000**	0.086±0.016	0.320±0.017	0.324±0.040	0.315±0.055	0.359±0.037	0.892±0.029	0.025±0.014	0.069±0.024
pid04	**1.000±0.000**	0.085±0.021	0.302±0.046	0.201±0.041	0.315±0.050	0.409±0.024	0.895±0.026	0.028±0.005	0.060±0.014
pid05	**0.973±0.024**	0.013±0.008	0.135±0.022	0.201±0.035	0.315±0.055	0.225±0.030	0.753±0.061	0.004±0.003	0.011±0.007
pid06	**0.943±0.019**	0.144±0.020	0.412±0.045	0.145±0.015	0.265±0.023	0.448±0.036	0.772±0.056	0.039±0.005	0.058±0.021
pid07	**0.742±0.036**	0.174±0.014	0.261±0.019	0.165±0.025	0.245±0.088	0.295±0.016	0.651±0.035	0.028±0.003	0.075±0.014
pid08	**0.996±0.011**	0.000±0.001	0.034±0.018	0.136±0.021	0.329±0.044	0.055±0.013	0.582±0.039	0.002±0.003	0.001±0.001
pid09	**0.990±0.012**	0.073±0.031	0.295±0.029	0.200±0.045	0.300±0.063	0.388±0.037	0.654±0.058	0.043±0.021	0.027±0.012
pid10	**1.000±0.000**	0.133±0.014	0.412±0.024	0.254±0.110	0.240±0.066	0.384±0.030	0.938±0.024	0.053±0.017	0.095±0.014
pid11	**1.000±0.000**	0.099±0.022	0.456±0.042	0.401±0.083	0.370±0.078	0.583±0.057	0.975±0.039	0.018±0.010	0.063±0.015
pid12	**1.000±0.000**	0.072±0.029	0.445±0.059	0.200±0.000	0.370±0.090	0.452±0.039	0.999±0.003	0.049±0.009	0.048±0.017
pid13	**0.991±0.027**	0.022±0.011	0.216±0.028	0.221±0.061	0.430±0.127	0.279±0.026	0.926±0.067	0.002±0.001	0.008±0.005
pid14	**0.962±0.038**	0.122±0.024	0.438±0.082	0.100±0.000	0.280±0.098	0.568±0.084	0.626±0.075	0.062±0.009	0.070±0.011
pid15	**0.642±0.072**	0.160±0.010	0.327±0.027	0.100±0.000	0.180±0.075	0.270±0.031	0.539±0.059	0.047±0.003	0.043±0.011
pid16	**1.000±0.000**	0.000±0.000	0.082±0.030	0.219±0.040	0.560±0.049	0.105±0.017	0.868±0.037	0.006±0.005	0.001±0.004
$D = 5$									
pid01	**0.833±0.046**	0.000±0.000	0.000±0.000	0.128±0.016	0.135±0.039	0.101±0.027	0.393±0.067	0.010±0.003	0.003±0.003
pid02	**0.825±0.037**	0.001±0.001	0.031±0.010	0.091±0.017	0.135±0.039	0.135±0.015	0.465±0.047	0.011±0.002	0.010±0.001
pid03	**0.766±0.045**	0.000±0.000	0.000±0.000	0.051±0.018	0.145±0.052	0.057±0.019	0.262±0.075	0.006±0.002	0.003±0.005
pid04	**0.647±0.059**	0.000±0.000	0.016±0.020	0.150±0.032	0.150±0.032	0.167±0.012	0.299±0.069	0.017±0.002	0.012±0.006
pid05	**0.383±0.056**	0.000±0.000	0.000±0.000	0.085±0.005	0.150±0.059	0.080±0.024	0.305±0.073	0.003±0.002	0.003±0.002
pid06	**0.440±0.058**	0.003±0.002	0.013±0.008	0.130±0.020	0.155±0.057	0.128±0.031	0.320±0.060	0.023±0.004	0.012±0.010
pid07	**0.270±0.047**	0.063±0.006	0.120±0.017	0.102±0.020	0.055±0.015	0.095±0.010	0.182±0.017	0.020±0.002	0.021±0.007
pid08	**0.599±0.073**	0.000±0.000	0.000±0.000	0.055±0.010	0.151±0.035	0.002±0.002	0.126±0.038	0.000±0.000	0.000±0.000
pid09	**0.899±0.042**	0.000±0.000	0.006±0.014	0.180±0.040	0.210±0.054	0.123±0.031	0.383±0.096	0.018±0.004	0.003±0.005
pid10	**0.892±0.039**	0.011±0.010	0.312±0.031	0.240±0.049	0.120±0.040	0.225±0.019	0.371±0.070	0.018±0.002	0.016±0.003
pid11	**0.894±0.045**	0.000±0.000	0.004±0.006	0.072±0.020	0.220±0.087	0.111±0.018	0.371±0.090	0.009±0.005	0.003±0.003
pid12	**0.861±0.112**	0.003±0.006	0.149±0.052	0.190±0.030	0.130±0.046	0.165±0.021	0.224±0.084	0.029±0.004	0.014±0.012
pid13	**0.546±0.129**	0.000±0.000	0.000±0.000	0.136±0.024	0.170±0.078	0.103±0.042	0.312±0.124	0.002±0.002	0.003±0.002
pid14	**0.398±0.076**	0.007±0.005	0.027±0.017	0.100±0.000	0.120±0.040	0.152±0.050	0.252±0.077	0.040±0.012	0.019±0.014
pid15	**0.329±0.039**	0.046±0.004	0.099±0.012	0.098±0.020	0.118±0.038	0.084±0.016	0.161±0.030	0.033±0.004	0.017±0.011
pid16	**0.590±0.102**	0.000±0.000	0.000±0.000	0.118±0.025	0.138±0.048	0.006±0.007	0.231±0.078	0.000±0.000	0.000±0.000
$D = 10$									
pid01	**0.638±0.045**	0.000±0.000	0.000±0.000	0.100±0.023	0.141±0.031	0.055±0.015	0.092±0.024	0.002±0.002	0.000±0.000
pid02	**0.418±0.048**	0.000±0.000	0.017±0.018	0.092±0.028	0.078±0.042	0.057±0.012	0.088±0.014	0.008±0.001	0.001±0.001
pid03	**0.570±0.045**	0.000±0.000	0.000±0.000	0.000±0.000	0.037±0.016	0.013±0.008	0.008±0.007	0.000±0.000	0.000±0.000
pid04	**0.490±0.041**	0.000±0.000	0.000±0.001	0.133±0.022	0.065±0.023	0.047±0.015	0.086±0.032	0.008±0.003	0.000±0.000
pid05	0.012±0.024	0.000±0.000	0.000±0.000	0.081±0.015	**0.101±0.032**	0.027±0.012	0.047±0.032	0.001±0.001	0.000±0.000
pid06	**0.139±0.027**	0.000±0.000	0.000±0.000	0.108±0.018	0.135±0.023	0.057±0.032	0.128±0.045	0.016±0.004	0.003±0.004
pid07	0.091±0.007	0.005±0.002	0.031±0.006	0.080±0.012	**0.096±0.030**	0.039±0.014	0.088±0.025	0.014±0.002	0.006±0.003
pid08	**0.255±0.070**	0.000±0.000	0.000±0.000	0.024±0.011	0.059±0.020	0.000±0.001	0.007±0.008	0.000±0.000	0.000±0.000
pid09	**0.675±0.047**	0.000±0.000	0.000±0.000	0.142±0.043	0.100±0.000	0.049±0.034	0.194±0.059	0.005±0.002	0.000±0.000
pid10	**0.582±0.059**	0.000±0.000	0.116±0.033	0.169±0.055	0.120±0.040	0.076±0.024	0.202±0.056	0.012±0.001	0.001±0.002
pid11	**0.477±0.045**	0.000±0.000	0.000±0.000	0.001±0.003	0.065±0.017	0.022±0.012	0.058±0.034	0.000±0.000	0.000±0.000
pid12	**0.669±0.119**	0.000±0.000	0.010±0.015	0.136±0.045	0.110±0.030	0.069±0.024	0.216±0.069	0.010±0.006	0.000±0.001
pid13	0.000±0.000	0.000±0.000	0.000±0.000	0.099±0.024	**0.127±0.039**	0.034±0.028	0.051±0.057	0.001±0.002	0.000±0.000
pid14	0.124±0.030	0.000±0.000	0.000±0.000	**0.156±0.051**	0.120±0.040	0.072±0.027	0.151±0.050	0.032±0.010	0.005±0.009
pid15	**0.151±0.004**	0.011±0.004	0.066±0.007	0.097±0.005	0.100±0.000	0.074±0.020	0.070±0.022	0.028±0.001	0.009±0.006
pid16	**0.355±0.105**	0.000±0.000	0.000±0.000	0.031±0.020	0.089±0.043	0.000±0.000	0.065±0.032	0.000±0.000	0.000±0.000
$D = 20$									
pid01	**0.500±0.032**	0.000±0.000	0.000±0.000	0.129±0.026	0.061±0.021	0.026±0.011	0.008±0.006	0.000±0.000	0.000±0.000
pid02	**0.325±0.000**	0.000±0.000	0.000±0.000	0.103±0.017	0.042±0.013	0.020±0.001	0.022±0.009	0.004±0.001	0.000±0.000
pid03	**0.458±0.073**	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.001	0.000±0.000	0.000±0.000	0.000±0.000
pid04	**0.127±0.027**	0.000±0.000	0.050±0.000	0.108±0.017	0.050±0.000	0.032±0.002	0.005±0.005	0.001±0.001	0.000±0.000
pid05	0.005±0.015	0.000±0.000	0.000±0.000	**0.082±0.013**	0.025±0.007	0.006±0.009	0.000±0.000	0.000±0.001	0.000±0.000
pid06	0.040±0.020	0.000±0.000	0.000±0.000	**0.126±0.022**	0.024±0.026	0.015±0.019	0.011±0.011	0.002±0.002	0.000±0.000
pid07	0.002±0.000	0.000±0.000	0.006±0.001	**0.040±0.008**	0.029±0.004	0.012±0.006	0.014±0.002	0.010±0.000	0.000±0.000
pid08	**0.294±0.054**	0.000±0.000	0.000±0.000	0.038±0.013	0.010±0.005	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000
pid09	**0.728±0.036**	0.000±0.000	0.000±0.000	0.181±0.027	0.113±0.033	0.037±0.021	0.105±0.055	0.000±0.000	0.000±0.000
pid10	**0.277±0.064**	0.000±0.000	0.030±0.010	0.230±0.046	0.085±0.028	0.039±0.002	0.076±0.008	0.006±0.001	0.000±0.000
pid11	**0.451±0.083**	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.001	0.000±0.000	0.000±0.000	0.000±0.000
pid12	**0.398±0.105**	0.000±0.000	0.016±0.014	0.220±0.040	0.100±0.000	0.065±0.004	0.078±0.030	0.003±0.002	0.000±0.000
pid13	0.014±0.042	0.000±0.000	0.000±0.000	**0.161±0.023**	0.064±0.007	0.023±0.019	0.021±0.033	0.000±0.000	0.000±0.000
pid14	0.083±0.000	0.000±0.000	0.000±0.000	**0.100±0.000**	0.097±0.008	0.018±0.036	0.086±0.031	0.021±0.008	0.000±0.000
pid15	**0.119±0.007**	0.000±0.000	0.004±0.003	0.083±0.012	0.079±0.005	0.049±0.008	0.053±0.017	0.019±0.001	0.000±0.000
pid16	**0.314±0.059**	0.000±0.000	0.000±0.000	0.062±0.023	0.026±0.010	0.000±0.000	0.005±0.013	0.000±0.000	0.000±0.000
Mean across 16 problems									
$D = 2$	**0.952±0.104**	0.084±0.057	0.287±0.131	0.205±0.089	0.318±0.111	0.337±0.144	0.791±0.148	0.029±0.021	0.047±0.032
$D = 5$	**0.636±0.226**	0.008±0.018	0.049±0.084	0.120±0.055	0.144±0.062	0.108±0.061	0.291±0.116	0.015±0.012	0.009±0.010
$D = 10$	**0.353±0.241**	0.001±0.003	0.015±0.033	0.091±0.059	0.096±0.041	0.043±0.031	0.097±0.074	0.009±0.010	0.002±0.004
$D = 20$	**0.259±0.213**	0.000±0.000	0.007±0.014	0.104±0.071	0.050±0.038	0.021±0.023	0.030±0.041	0.004±0.007	0.000±0.000

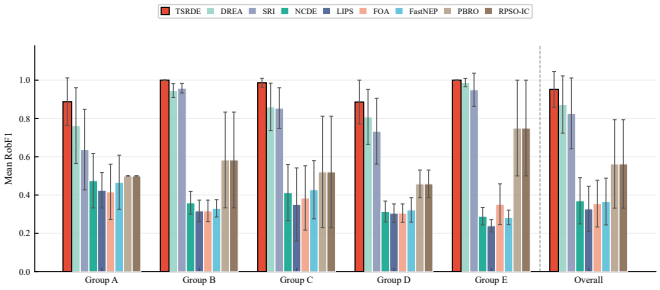


Fig. S-2. Mean RobF1 by problem group for all nine algorithms (error bars: ±std across the four per-group problems), summarizing Table S-III and the group averages of Table III in the main paper. TSRDE (leftmost, dark-edged bar in each cluster) leads every group; the margin over the two closest baselines, DREA and SRI, is largest on the anisotropic Group D.

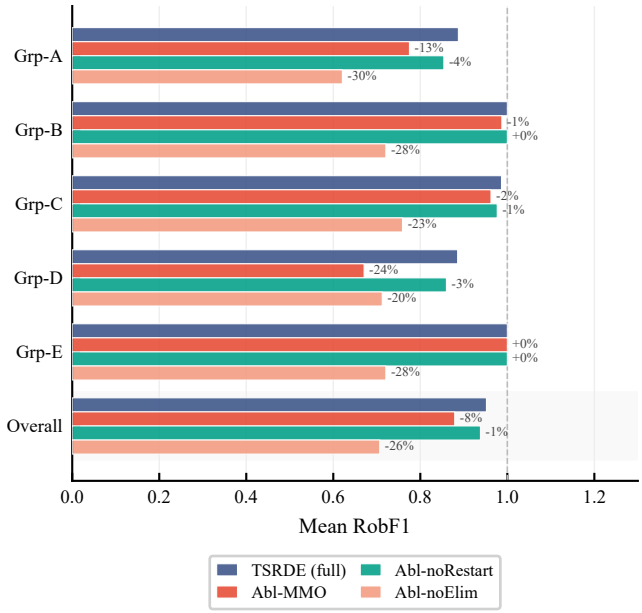


Fig. S-3. Ablation study of the main paper (Section V-D): mean RobF1 per problem group for the full TSRDE and its three ablated variants, all sharing the same codebase and seeds. Abl-noElim is uniformly damaging across all five groups; Abl-MMO is most damaging on Group D; Abl-noRestart has the smallest overall effect.

TABLE S-VII
NEAR-OPTIMALITY AUDIT OF ALL RETURNED SOLUTIONS (200 RUNS PER ALGORITHM, 10 RUNS × 20 PROBLEMS). "PASS RATE" IS THE FRACTION OF SOLUTIONS WITH $f(x) - f^* < 10^{-2}$.

Algorithm	n **sol.**	**Median** $f-f^*$	**Max** $f-f^*$	**Pass rate**	**In-niche**
TSRDE	385	0	4.6×10^{-14}	1.000	1.000
NCDE	1000	0	4.7×10^{-12}	1.000	1.000
LIPS	1000	0	1.6×10^{-12}	1.000	1.000
PBRO	410	2.1×10^{-3}	3.0×10^{-2}	0.946	1.000
RPSO-IC	410	3.3×10^{-3}	8.2	0.700	1.000
SRI	410	1.8×10^{-15}	4.4	0.939	1.000
FOA	1000	5.9×10^{-4}	8.4×10^{-2}	0.922	1.000
DREA	410	2.3×10^{-1}	5.5	0.032	1.000
FastNEP	1000	1.3×10^{0}	6.2	0.254	1.000

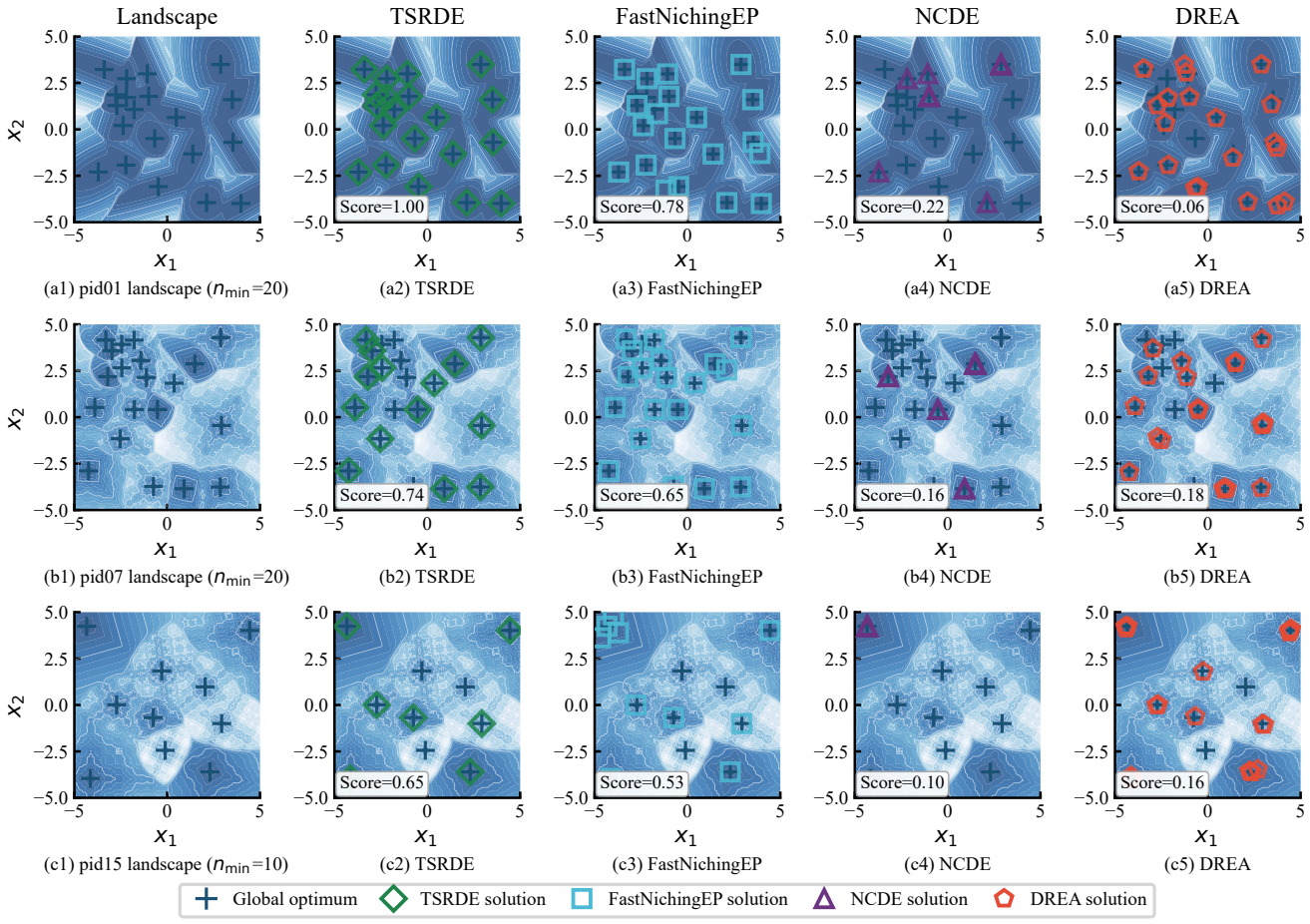


Fig. S-4. Solution distributions on three representative CEC2026 problems at $D = 2$ (rows, increasing difficulty top to bottom), using the run closest to each algorithm's own 10-run median score. Crosses mark ground-truth global optima. Columns show TSRDE, FastNEP, NCDE, and DREA. TSRDE aligns closely with optima across all three difficulty levels; FastNEP and NCDE show increasingly sparse coverage at higher difficulty; DREA solutions are widely scattered and miss most peaks.

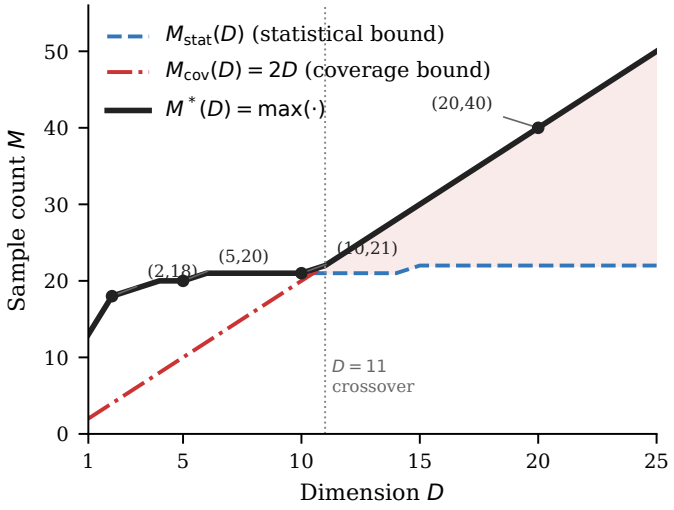


Fig. S-5. Dimension-adaptive sample count $M^*(D)$. The statistical bound M_{stat} (blue dashed) increases slowly with dimension due to the decreasing excess kurtosis of the perturbed-objective distribution; the directional coverage bound $M_{\text{cov}} = 2D$ (red dash-dot) grows linearly. $M^*(D) = \max(M_{\text{stat}}, M_{\text{cov}})$ (black solid) remains nearly flat through $D \approx 13$ and then increases linearly, balancing statistical reliability against directional coverage.

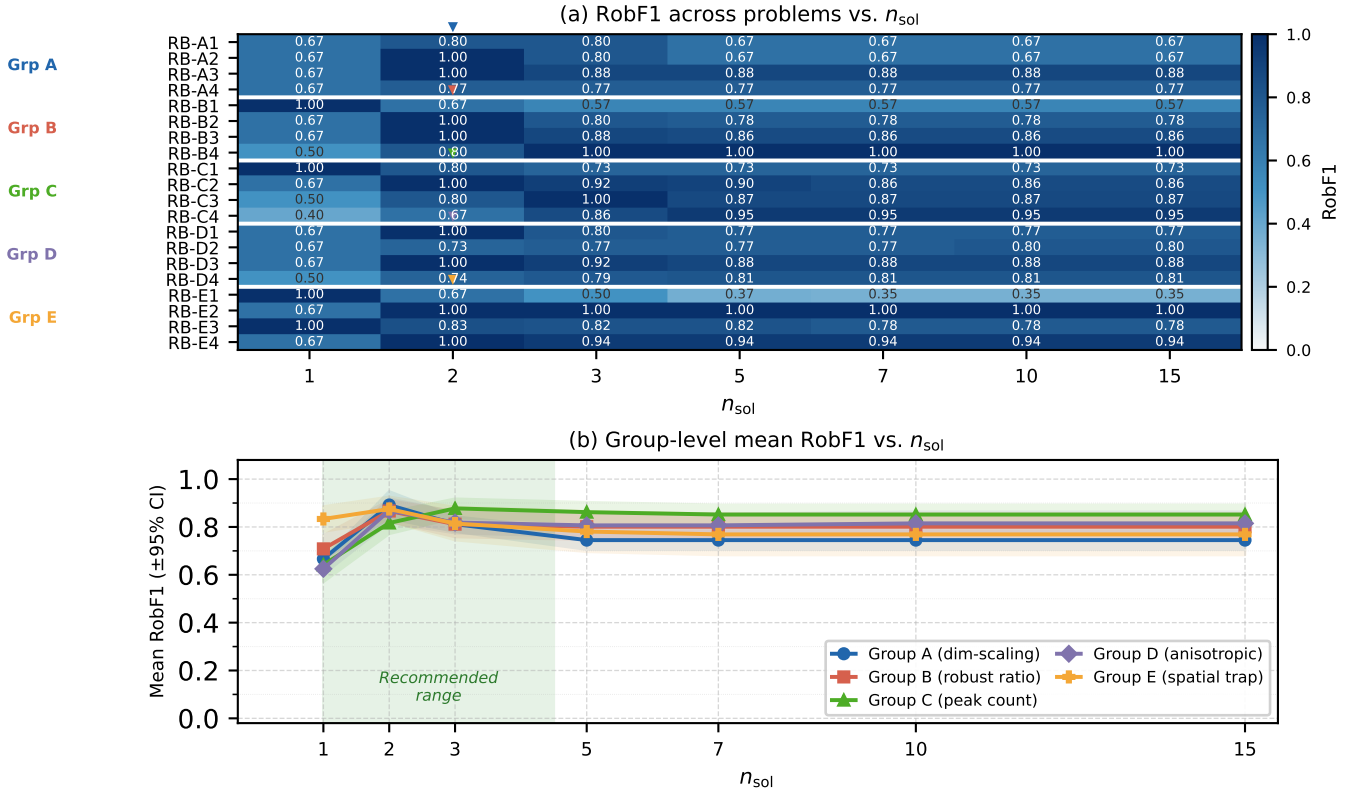


Fig. S-6. RobF1 versus n_{sol} for each problem group. Top: heatmap over all 20 problems at seven tested values. Bottom: group-level mean with 95% confidence intervals. The shaded grey band marks the tolerance region $n_{\mathrm{sol}} \in [n_{\mathrm{rob}}/2,\ 2n_{\mathrm{rob}}]$ for a representative $n_{\mathrm{rob}} = 2$.